

Ex no:6

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Experiment 6: Dimensionality Reduction and Model Evaluation (With and Without PCA)

AIM

To analyze the impact of dimensionality reduction using Principal Component Analysis (PCA) on the performance and stability of various machine learning classifiers by comparing models trained with and without PCA using 5-fold cross-validation.

OBJECTIVE

- To preprocess and standardize the dataset before model training.
- To apply PCA for dimensionality reduction while retaining 95% variance.
- To train multiple classifiers in the original feature space (No-PCA).
- To train the same classifiers in the reduced feature space (With-PCA).
- To evaluate performance using accuracy, F1-score, and 5-fold cross-validation.
- To compare stability and overfitting behavior across models.

Support Vector Machine (SVM), Naive Bayes, k-Nearest Neighbors (KNN)

Dataset used

- Spambase dataset
- From assignment-3

```
In [2]: import pandas as pd
#LOADING THE DATASET
df=pd.read_csv("spambase_csv.csv")
print("Number of Samples :", df.shape[0])
print("Number of Features :", df.shape[1] - 1)
```

Number of Samples : 4601
Number of Features : 57

```
In [3]: print(df.columns)
print(df.shape)
```

```
Index(['word_freq_make', 'word_freq_address', 'word_freq_all', 'word_freq_3d',
      'word_freq_our', 'word_freq_over', 'word_freq_remove',
      'word_freq_internet', 'word_freq_order', 'word_freq_mail',
      'word_freq_receive', 'word_freq_will', 'word_freq_people',
      'word_freq_report', 'word_freq_addresses', 'word_freq_free',
      'word_freq_business', 'word_freq_email', 'word_freq_you',
      'word_freq_credit', 'word_freq_your', 'word_freq_font', 'word_freq_000',
      'word_freq_money', 'word_freq_hp', 'word_freq_hpl', 'word_freq_george',
      'word_freq_650', 'word_freq_lab', 'word_freq_labs', 'word_freq_telnet',
      'word_freq_857', 'word_freq_data', 'word_freq_415', 'word_freq_85',
      'word_freq_technology', 'word_freq_1999', 'word_freq_parts',
      'word_freq_pm', 'word_freq_direct', 'word_freq_cs', 'word_freq_meeting',
      'word_freq_original', 'word_freq_project', 'word_freq_re',
      'word_freq_edu', 'word_freq_table', 'word_freq_conference',
      'char_freq_%3B', 'char_freq_%28', 'char_freq_%5B', 'char_freq_%21',
      'char_freq_%24', 'char_freq_%23', 'capital_run_length_average',
      'capital_run_length_longest', 'capital_run_length_total', 'class'],
      dtype='object')
(4601, 58)
```

```
In [4]: #STATISTICS
df.describe()
```

Out [4]:

	word_freq_make	word_freq_address	word_freq_all	word_freq_3d	word_freq_our	word_freq_over	word_freq_remove	wc
count	4601.000000	4601.000000	4601.000000	4601.000000	4601.000000	4601.000000	4601.000000	
mean	0.104553	0.213015	0.280656	0.065425	0.312223	0.095901	0.114208	
std	0.305358	1.290575	0.504143	1.395151	0.672513	0.273824	0.391441	
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	
50%	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	
75%	0.000000	0.000000	0.420000	0.000000	0.380000	0.000000	0.000000	
max	4.540000	14.280000	5.100000	42.810000	10.000000	5.880000	7.270000	

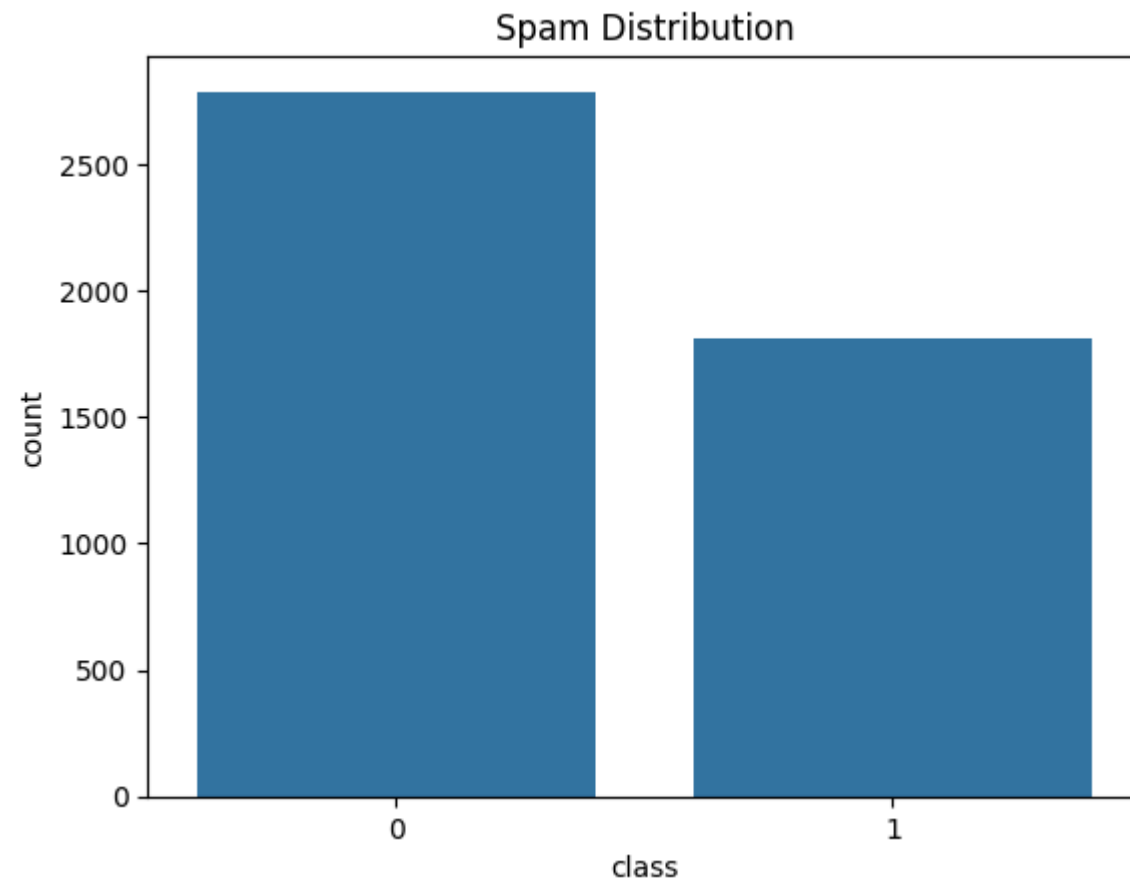
8 rows × 8 columns

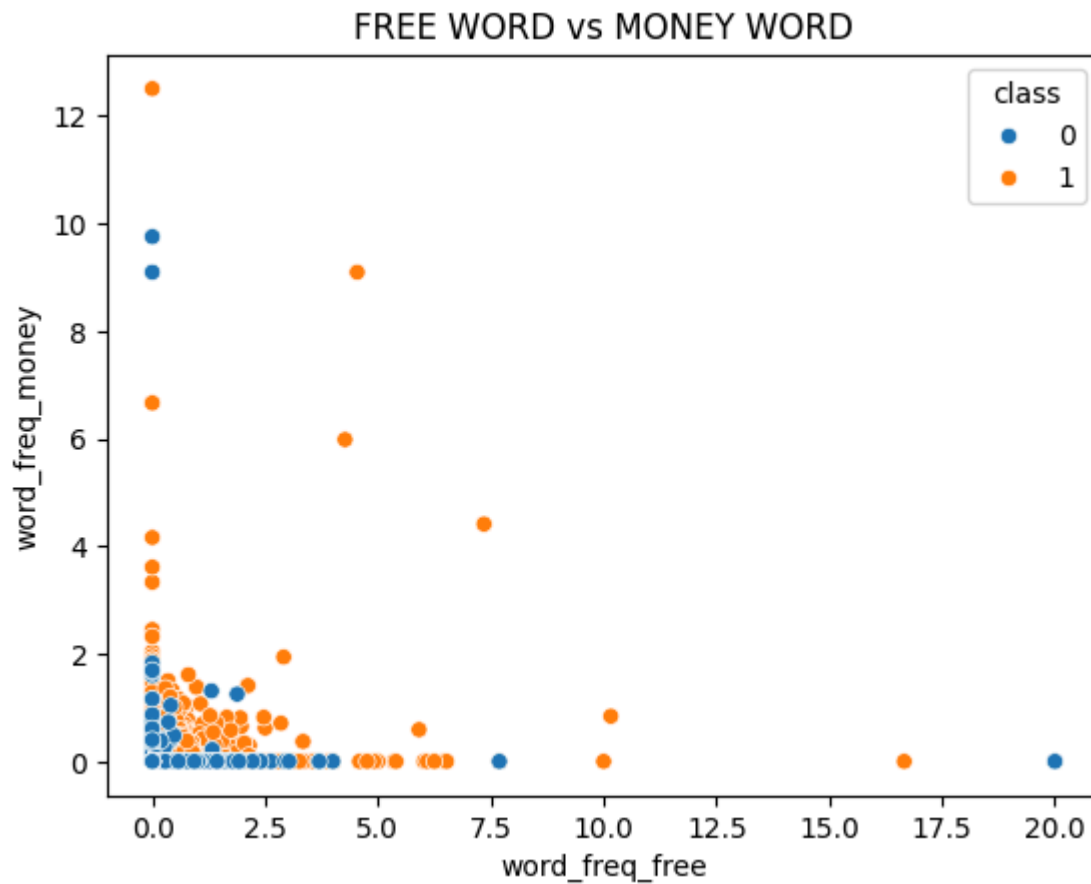
```
In [5]: #FEATURES
x=df.drop(["class","capital_run_length_average","capital_run_length_longest","capital_run_length_total"],axis=1)
y=df["class"]

#Normalizing or standardizing the features
from sklearn.preprocessing import StandardScaler,MinMaxScaler
scaler=StandardScaler()
x_scaled=scaler.fit_transform(x)
```

```
In [6]: import matplotlib.pyplot as plt
import seaborn as sns
sns.countplot(x="class",data=df)
plt.title("Spam Distribution")
plt.show()

sns.scatterplot(data=df,x="word_freq_free",y="word_freq_money",hue="class")
plt.title("FREE WORD vs MONEY WORD")
plt.show
plt.show()
```





In [7]: *#SPLITTING DATA INTO TRAIN SET AND TEST SET*

```
from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x_scaled, y, test_size=0.2, random_state=42)
```

In [8]: **from** sklearn.svm **import** SVC

from sklearn.naive_bayes **import** GaussianNB

import numpy **as** np

from sklearn.neighbors **import** KNeighborsClassifier

```
models={"SVM":(SVC(probability=True),{"kernel":["linear","rbf"],"C":[0.1,1,10],"gamma":["scale","auto"]}),
        "Naive Bayes":(GaussianNB(),{"var_smoothing":np.logspace(0,-9,num=100)}),
        "KNN":(KNeighborsClassifier(),{"n_neighbors":[3,5,7],"weights":["uniform","distance"],"metric":["euclidean"]})
}
```

```
In [9]: from sklearn.model_selection import GridSearchCV, cross_val_score
from sklearn.metrics import accuracy_score, f1_score

#without pca

def evaluate(name, m, params, x_tr, x_te):
    if params:
        grid = GridSearchCV(m, params, cv=5, scoring="accuracy")
        grid.fit(x_tr, y_train)
        best = grid.best_estimator_
    else:
        best = m.fit(x_tr, y_train)
        y_pred = best.predict(x_te)
        acc = accuracy_score(y_test, y_pred)
        f1 = f1_score(y_test, y_pred, average="weighted")
        cv_scores = cross_val_score(best, x_tr, y_train, cv=5)
        return acc, f1, cv_scores, best
result_no_pca = {}
best_models_no_pca = {}
for name, (m, param) in models.items():
    acc, f1, cv, best = evaluate(name, m, param, x_train, x_test)
    result_no_pca[name] = (acc, f1, cv)
    best_models_no_pca[name] = best
```

```
In [10]: print("PERFORMANCE WITHOUT PCA")
df1 = pd.DataFrame.from_dict(result_no_pca, orient='index', columns=['Accuracy', 'F1-Score', 'CV Score'])
folds_df = df1['CV Score'].apply(pd.Series)
folds_df.columns = [f'Fold {i+1}' for i in range(folds_df.shape[1])]
df_final = pd.concat([df1.drop(columns=['CV Score']), folds_df], axis=1)
df_final
```

PERFORMANCE WITHOUT PCA

```
Out[10]:
```

	Accuracy	F1-Score	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
SVM	0.934853	0.934477	0.934783	0.902174	0.934783	0.941576	0.932065
Naive Bayes	0.815418	0.815923	0.819293	0.790761	0.808424	0.824728	0.823370
KNN	0.918567	0.917776	0.927989	0.903533	0.907609	0.932065	0.919837

```
In [11]: from sklearn.decomposition import PCA
pca=PCA(n_components=0.95)
x_train_pca=pca.fit_transform(x_train)
x_test_pca=pca.transform(x_test)

n_components = pca.n_components_
explained_variance = pca.explained_variance_ratio_.sum() * 100

print("Chosen Components:", n_components)
print("Explained Variance:", explained_variance)
print("ORIGINAL SHAPE:",x_train.shape)

print("REDUCED SHAPE:",x_train_pca.shape)
```

```
Chosen Components: 45
Explained Variance: 95.00850406937306
ORIGINAL SHAPE: (3680, 54)
REDUCED SHAPE: (3680, 45)
```

```

/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:604: RuntimeWarning: divide by
zero encountered in matmul
    C = X.T @ X
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:604: RuntimeWarning: overflow
encountered in matmul
    C = X.T @ X
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:604: RuntimeWarning: invalid v
alue encountered in matmul
    C = X.T @ X
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: divide b
y zero encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: overflow
encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: invalid
value encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: divide b
y zero encountered in matmul
    X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: overflow
encountered in matmul
    X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: invalid
value encountered in matmul
    X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: divide b
y zero encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: overflow
encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: invalid
value encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: divide b
y zero encountered in matmul
    X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: overflow

```

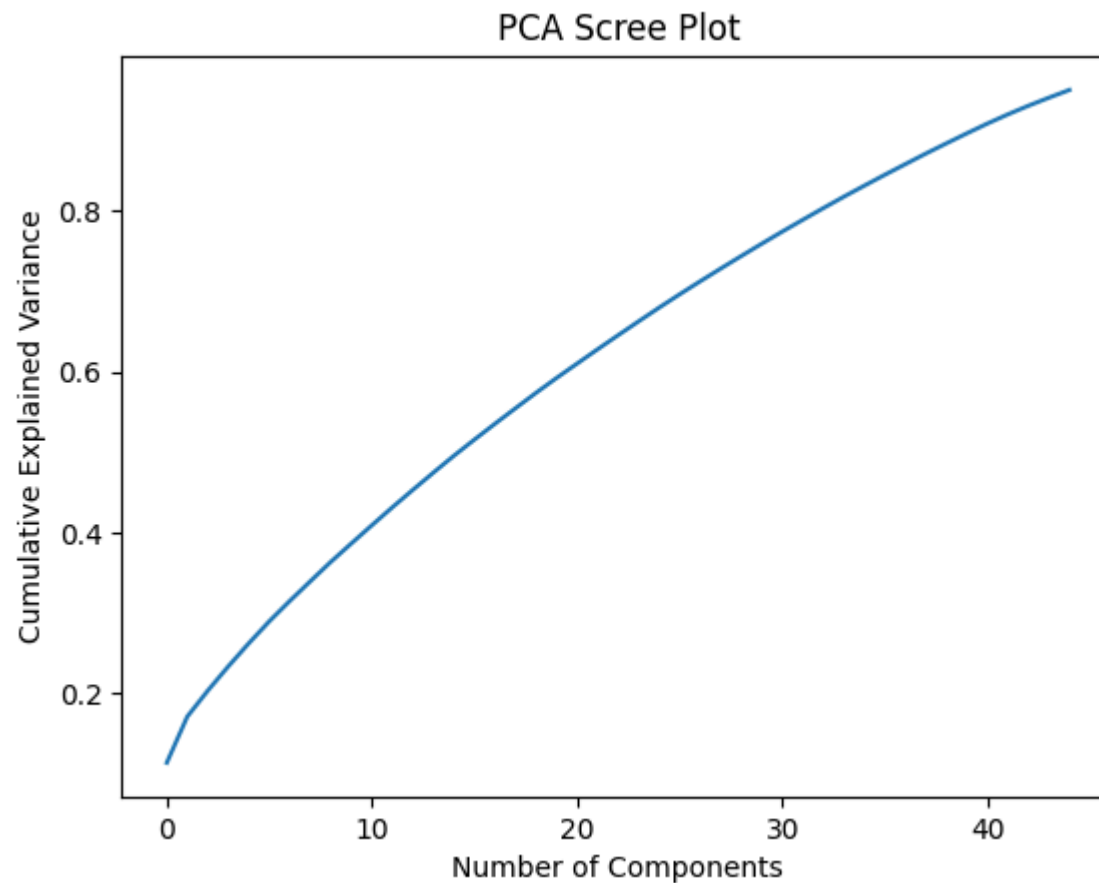

encountered in matmul

```
X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
```

/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: invalid value encountered in matmul

```
X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
```

```
In [12]: #Scree plot
plt.plot(np.cumsum(pca.explained_variance_ratio_))
plt.xlabel("Number of Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("PCA Scree Plot")
plt.show()
```



```
In [13]: pca_table=pd.DataFrame({"Settings":["With-PCA"],"Chosen Components":[f"{n_components}"],"Explained Variance (%)":[f
pca_table
```

```
Out[13]:
```

	Settings	Chosen Components	Explained Variance (%)	Justification
0	With-PCA	45	95.00850406937306	Dimensionality reduced while retaining most da...

```
In [16]: #with pca
result_pca = {}
best_models_pca = {}
for name, (m, param) in models.items():
    acc, f1, cv, best = evaluate(name, m, param,x_train_pca, x_test_pca)
    result_pca[name] = (acc, f1, cv)
    best_models_pca[name] = best
```

```
In [17]: print("PERFORMANCE WITHOUT PCA")
df2 = pd.DataFrame.from_dict(result_pca, orient='index', columns=['Accuracy', 'F1-Score', 'CV Score'])
folds_df = df2['CV Score'].apply(pd.Series)
folds_df.columns = [f'Fold {i+1}' for i in range(folds_df.shape[1])]
df_final = pd.concat([df2.drop(columns=['CV Score']), folds_df], axis=1)
df_final
```

PERFORMANCE WITHOUT PCA

```
Out[17]:
```

	Accuracy	F1-Score	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
SVM	0.931596	0.931218	0.932065	0.903533	0.932065	0.938859	0.927989
Naive Bayes	0.805646	0.805710	0.828804	0.760870	0.864130	0.701087	0.817935
KNN	0.906623	0.906173	0.918478	0.914402	0.925272	0.917120	0.908967

```
In [18]: print("SVM - Hyperparameter Tuning Results")
svm_table = pd.DataFrame({"Kernel": ["linear / rbf"],"C Values Tried": ["0.1, 1, 10"],"Gamma Values Tried": ["scale
svm_table
```

SVM - Hyperparameter Tuning Results

Out [18]:

	Kernel	C Values Tried	Gamma Values Tried	Performance (No PCA)	Performance (With PCA)
0	linear / rbf	0.1, 1, 10	scale, auto	0.934853	0.931596

```
In [19]: print("Naive Bayes – Smoothing Choices")
nb_table = pd.DataFrame({"Smoothing Parameter": ["logspace(0, -9)"],"Performance (No PCA)": [result_no_pca["Naive B
nb_table
```

Naive Bayes – Smoothing Choices

Out [19]:

	Smoothing Parameter	Performance (No PCA)	Performance (With PCA)
0	logspace(0, -9)	0.815418	0.805646

```
In [20]: print("KNN – Hyperparameter Tuning")
knn_table = pd.DataFrame({"k Values": ["3, 5, 7"],"Weights": ["uniform, distance"],"Distance Metrics": ["euclidean,
knn_table
```

KNN – Hyperparameter Tuning

Out [20]:

	k Values	Weights	Distance Metrics	Performance (No PCA)	Performance (With PCA)
0	3, 5, 7	uniform, distance	euclidean, manhattan	0.918567	0.906623

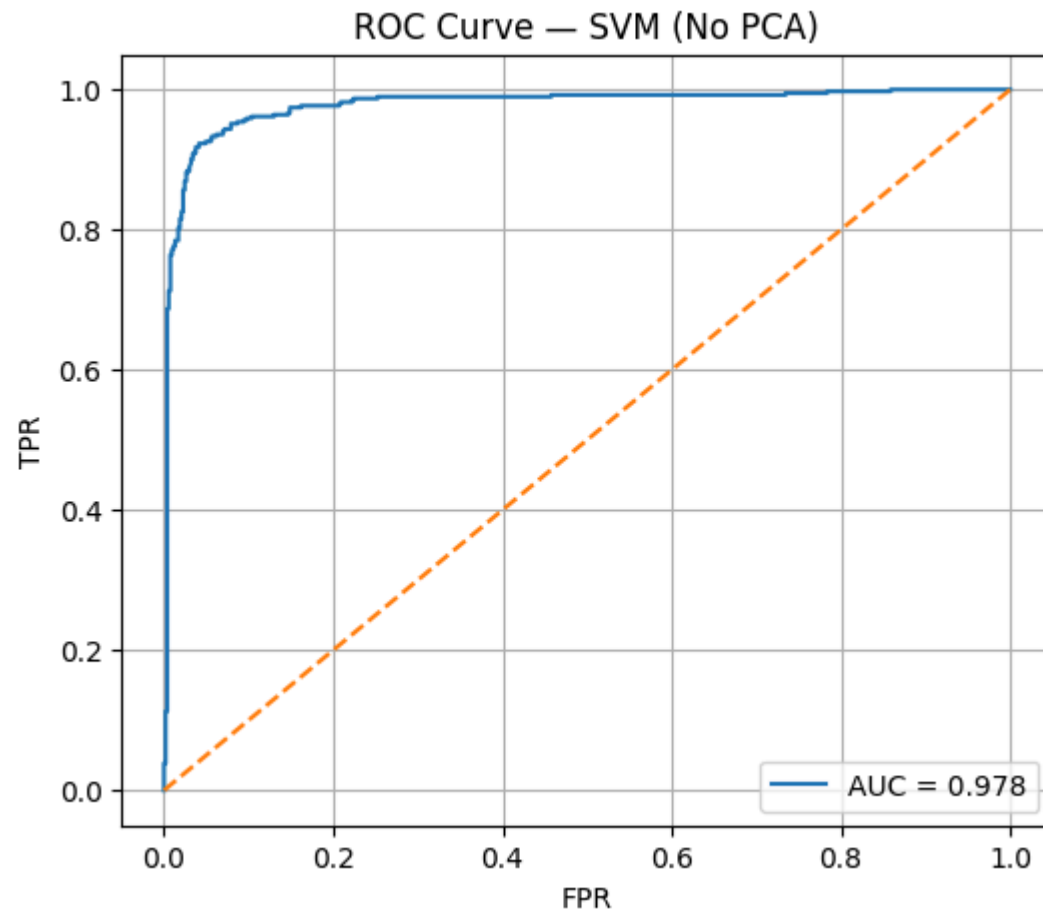
```
In [21]: from sklearn.metrics import roc_curve, auc, confusion_matrix, ConfusionMatrixDisplay
import matplotlib.pyplot as plt

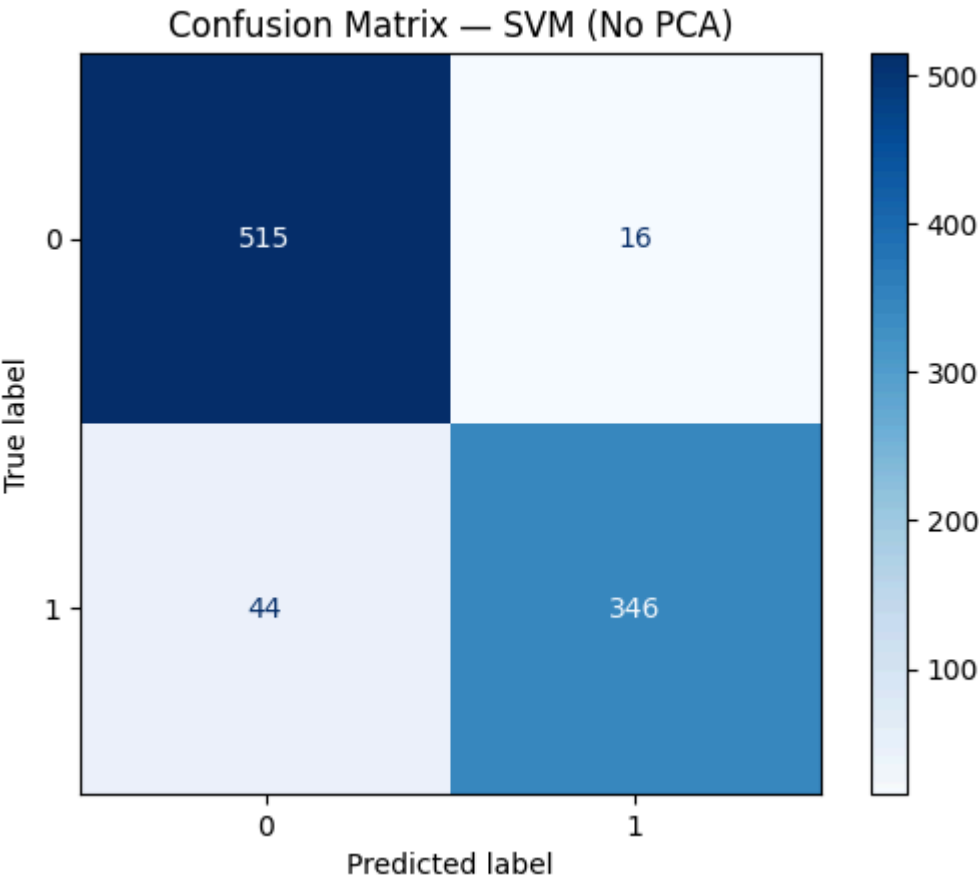
def plot_roc_cm(model, X_test_used, y_test, title):
    y_prob = model.predict_proba(X_test_used)[: , 1]
    # ROC
    fpr, tpr, _ = roc_curve(y_test, y_prob)
    roc_auc = auc(fpr, tpr)
    plt.figure(figsize=(6,5))
    plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
    plt.plot([0,1],[0,1], '--')
    plt.title(f"ROC Curve – {title}")
    plt.xlabel("FPR")
    plt.ylabel("TPR")
    plt.legend()
    plt.grid(True)
```

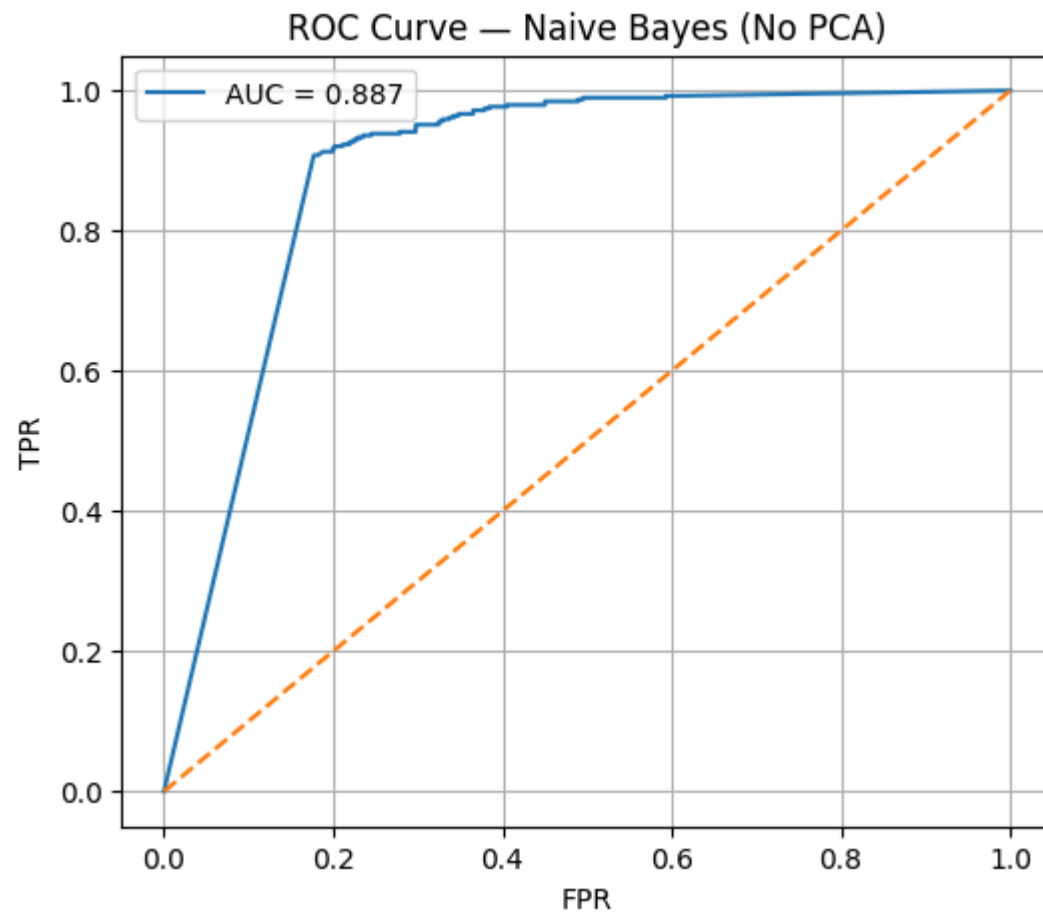
```
plt.show()

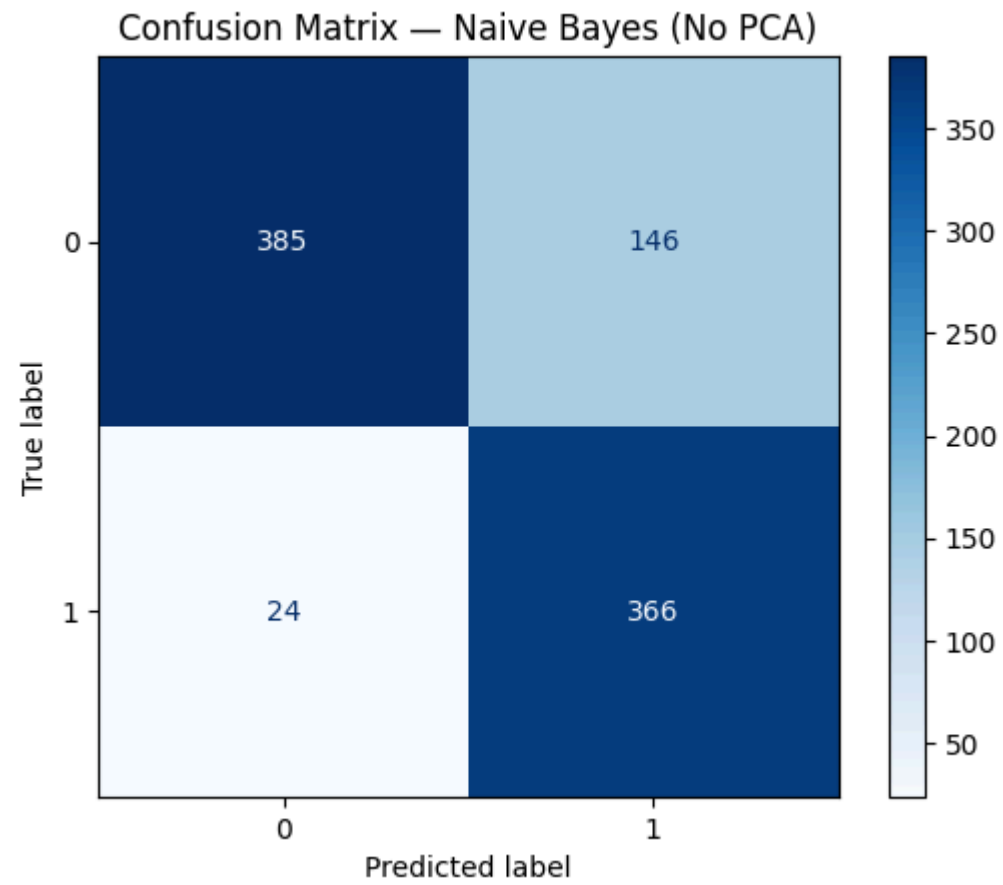
# Confusion Matrix
y_pred = model.predict(X_test_used)
cm = confusion_matrix(y_test, y_pred)
ConfusionMatrixDisplay(cm).plot(cmap="Blues")
plt.title(f"Confusion Matrix - {title}")
plt.show()
```

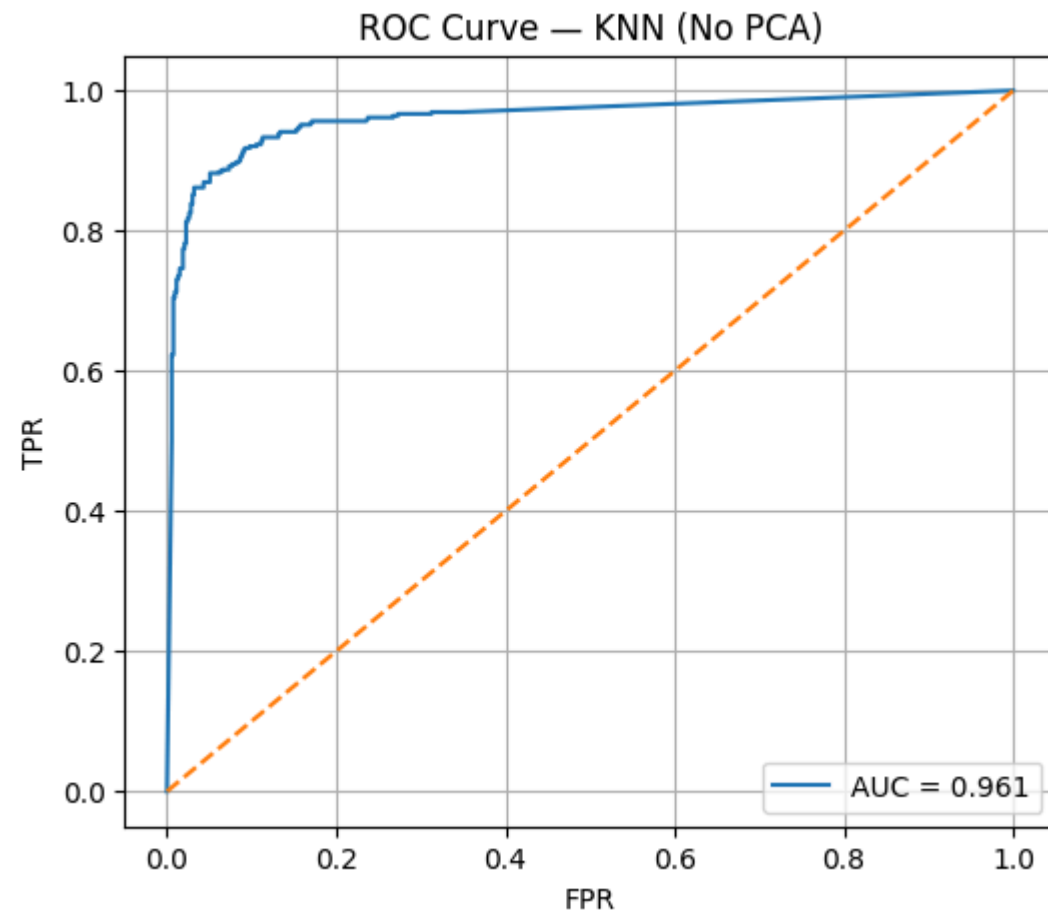
```
In [22]: plot_roc_cm(best_models_no_pca["SVM"], x_test, y_test, "SVM (No PCA)")
plot_roc_cm(best_models_no_pca["Naive Bayes"], x_test, y_test, "Naive Bayes (No PCA)")
plot_roc_cm(best_models_no_pca["KNN"], x_test, y_test, "KNN (No PCA)")
```

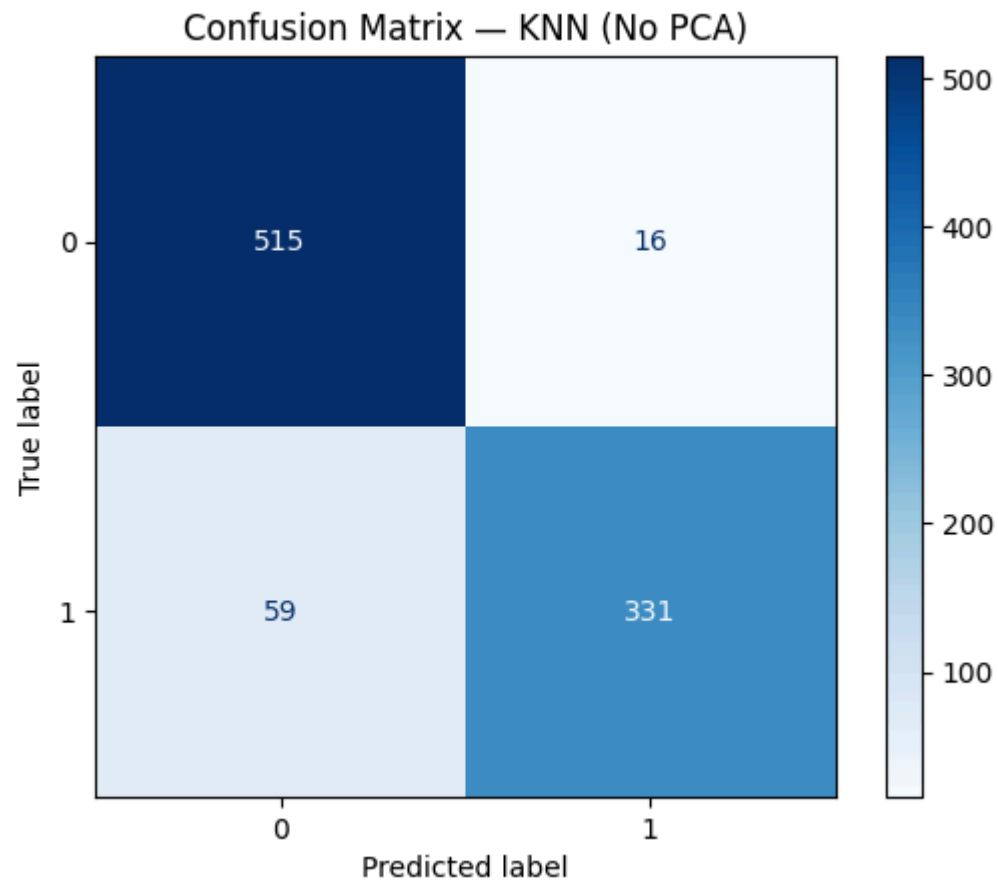




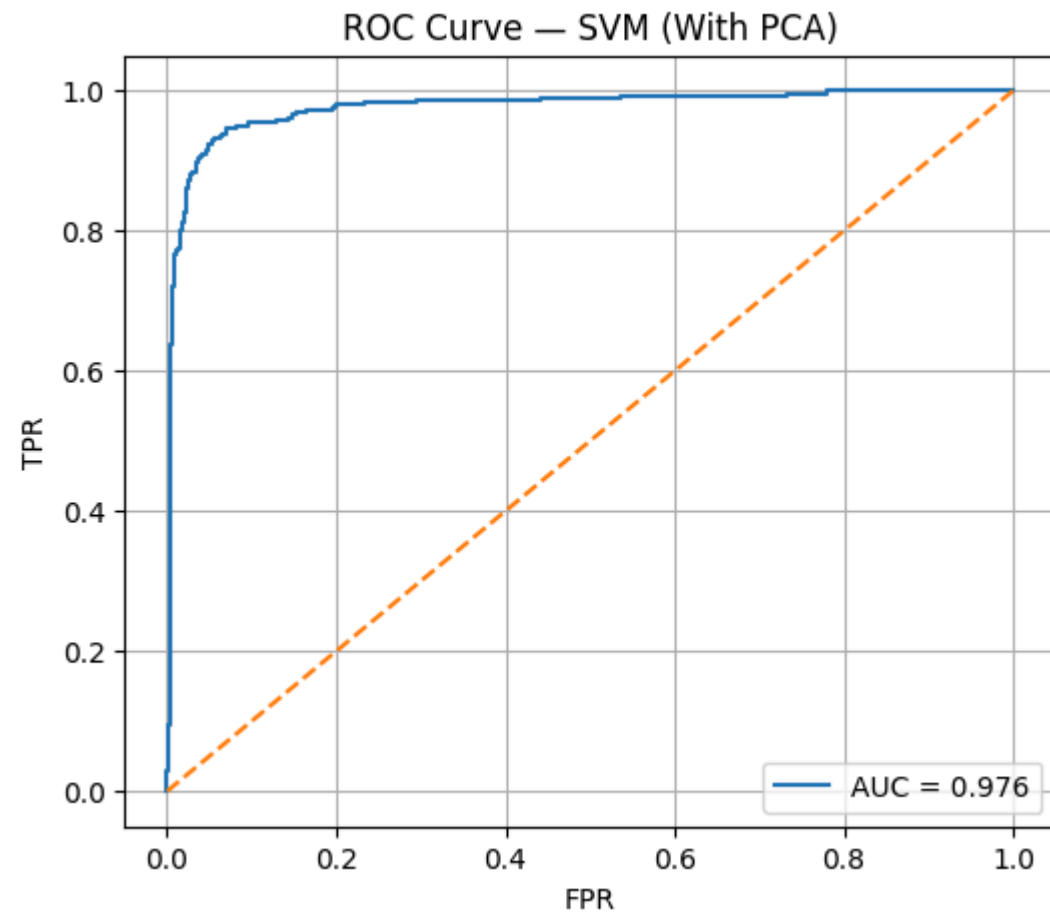


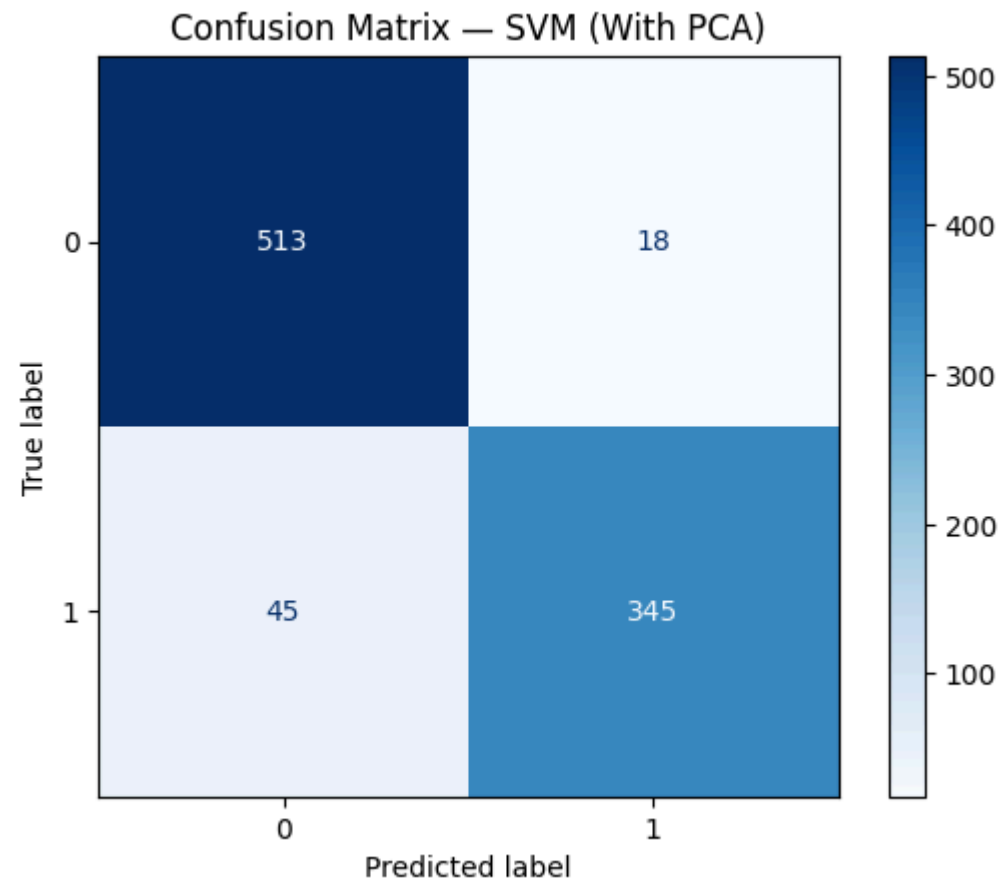


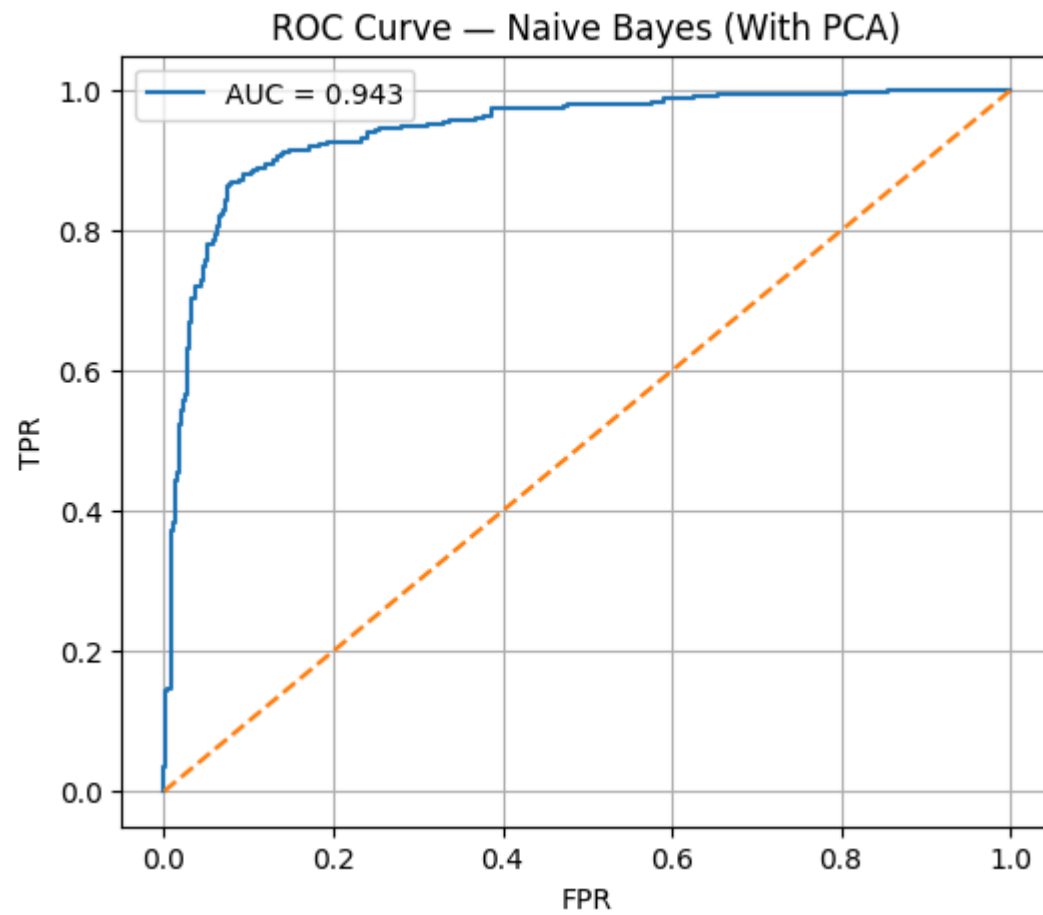


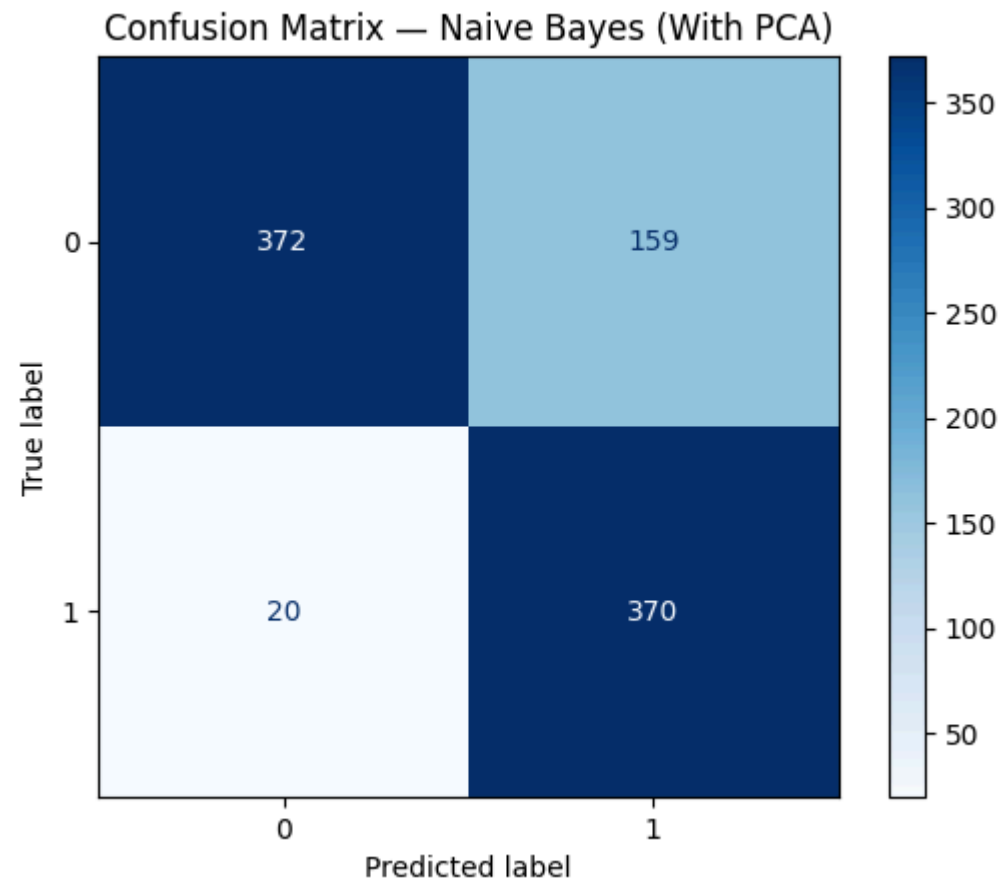


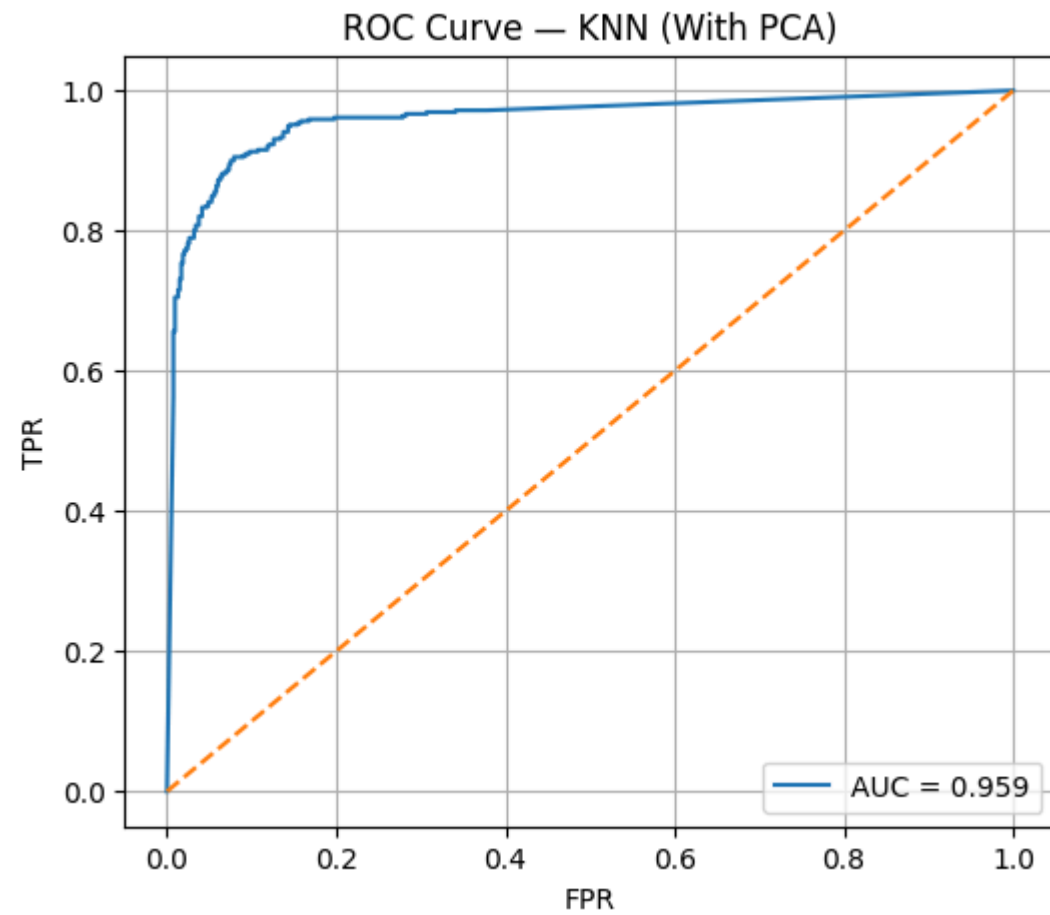
```
In [23]: plot_roc_cm(best_models_pca["SVM"], x_test_pca, y_test, "SVM (With PCA)")  
plot_roc_cm(best_models_pca["Naive Bayes"], x_test_pca, y_test, "Naive Bayes (With PCA)")  
plot_roc_cm(best_models_pca["KNN"], x_test_pca, y_test, "KNN (With PCA)")
```

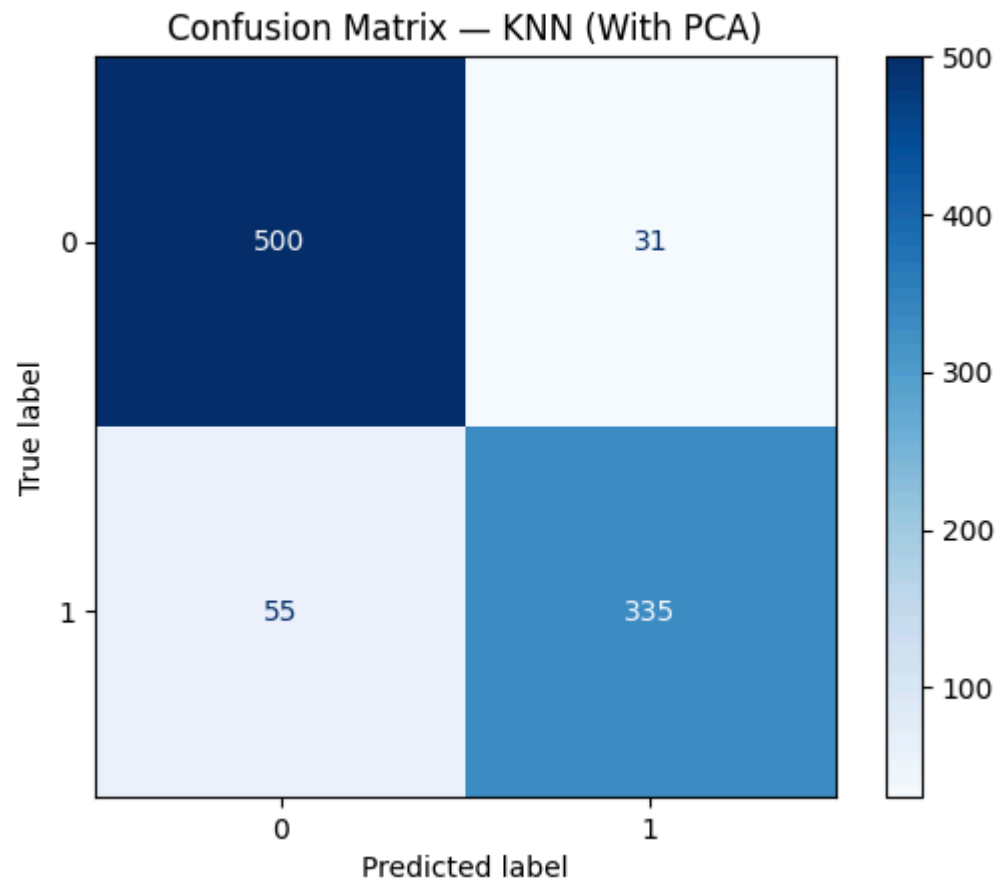












Logistic Regression

Dataset used

- Loan prediction dataset
- From assignment-2

```
In [24]: df_log=pd.read_csv("train.csv")
print("Number of Samples :", df_log.shape[0])
print("Number of Features :", df_log.shape[1] - 1)
```

Number of Samples : 30000

Number of Features : 23

```
In [25]: #STATISTICS
df_log.describe()
```

Out[25]:

	Age	Income (USD)	Loan Amount Request (USD)	Current Loan Expenses (USD)	Dependents	Credit Score	No. of Defaults	Property ID	P
count	30000.000000	2.542400e+04	30000.000000	29828.000000	27507.000000	28297.000000	30000.000000	30000.000000	2.
mean	40.092300	2.630574e+03	88826.333855	400.936876	2.253027	739.885381	0.193933	501.934700	2.
std	16.045129	1.126272e+04	59536.949605	242.545375	0.951162	72.163846	0.395384	288.158086	1.
min	18.000000	3.777000e+02	6048.240000	-999.000000	1.000000	580.000000	0.000000	1.000000	3.
25%	25.000000	1.650457e+03	41177.755000	247.667500	2.000000	681.880000	0.000000	251.000000	1.
50%	40.000000	2.222435e+03	75128.075000	375.205000	2.000000	739.820000	0.000000	504.000000	2.
75%	55.000000	3.090593e+03	119964.605000	521.292500	3.000000	799.120000	0.000000	751.000000	3.
max	65.000000	1.777460e+06	621497.820000	3840.880000	14.000000	896.260000	1.000000	999.000000	1.

```
In [26]: #dropping unessential columns
df_log=df_log.drop("Customer ID",axis=1)
df_log=df_log.drop("Name",axis=1)
```

```
In [27]: # Handling missing values
for col in df_log.columns:
    if df_log[col].dtype=='object':
        df_log[col]=df_log[col].fillna(df_log[col].mode()[0])
    else:
        df_log[col]=df_log[col].fillna(df_log[col].median())
```

```
In [28]: #one hot encoding
df_log=pd.get_dummies(df_log,drop_first=True)
```



```

In [29]: #Outlier Handling
df_log = df_log[df_log["Income (USD)"] < df_log["Income (USD)"].quantile(0.99)]
df_log = df_log[df_log["Property Price"] < df_log["Property Price"].quantile(0.99)]

df_log["log_income"] = np.log1p(df_log["Income (USD)"])
median=df_log.loc[df_log["Current Loan Expenses (USD)"] >=0,"Current Loan Expenses (USD)"].median()
df_log.loc[df_log["Current Loan Expenses (USD)"]<0,"Current Loan Expenses (USD)"]=median
df_log["log_expense"] = np.log1p(df_log["Current Loan Expenses (USD)"])

In [30]: df_log['Loan Sanction Amount (USD)']=df_log['Loan Sanction Amount (USD)'].fillna(df_log['Loan Sanction Amount (USD)']
(df_log["Loan Sanction Amount (USD)"]<0).sum()
median=df_log.loc[df_log["Loan Sanction Amount (USD)"] >=0,"Loan Sanction Amount (USD)"].median()
df_log.loc[df_log["Loan Sanction Amount (USD)"]<0,"Loan Sanction Amount (USD)"]=median
df_log['log_loan_amount'] = np.log1p(df_log['Loan Sanction Amount (USD)'])

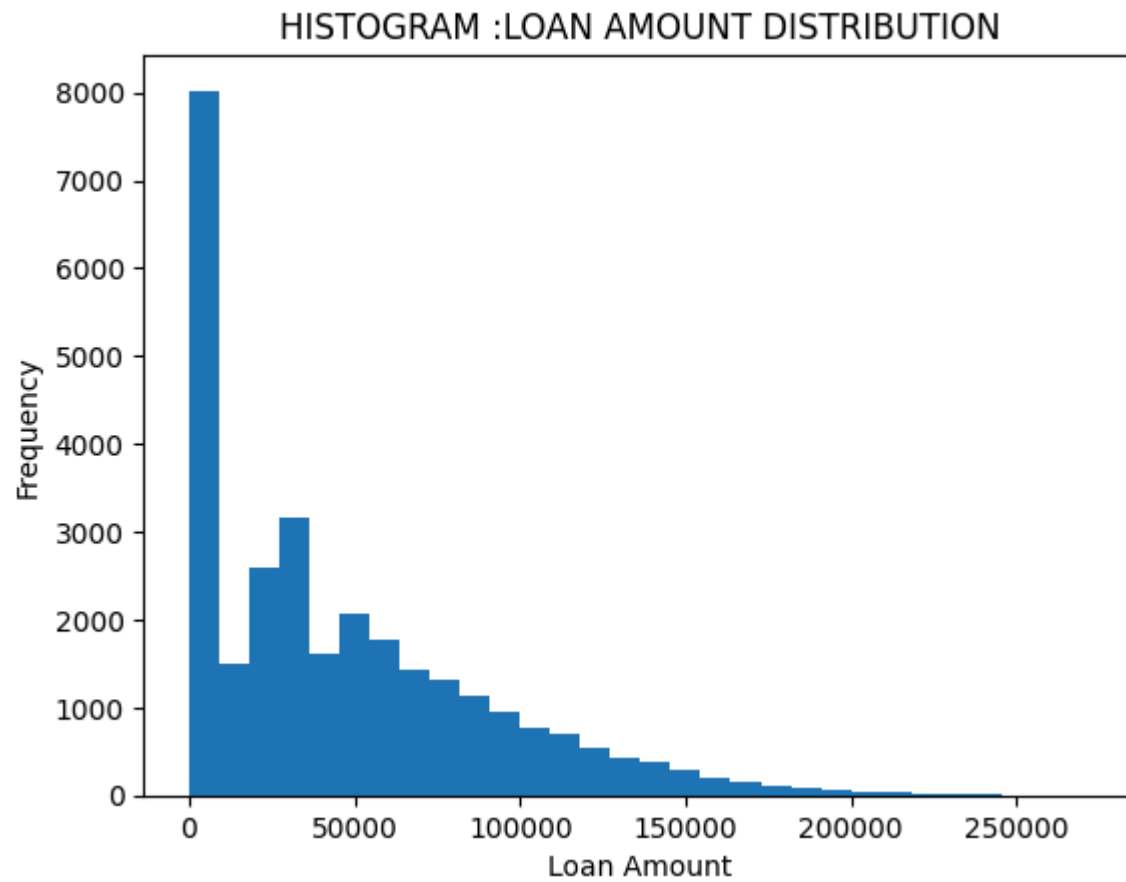
In [31]: #Adding Ratio-Based Features
df_log["Loan_to_Income"] = df_log["Loan Amount Request (USD)"] / (df_log["Income (USD)"] + 1)

In [32]: #selecting features(x,y)
x=df_log.drop(["Loan Sanction Amount (USD)"],axis=1)
y=df_log["Loan Sanction Amount (USD)"]

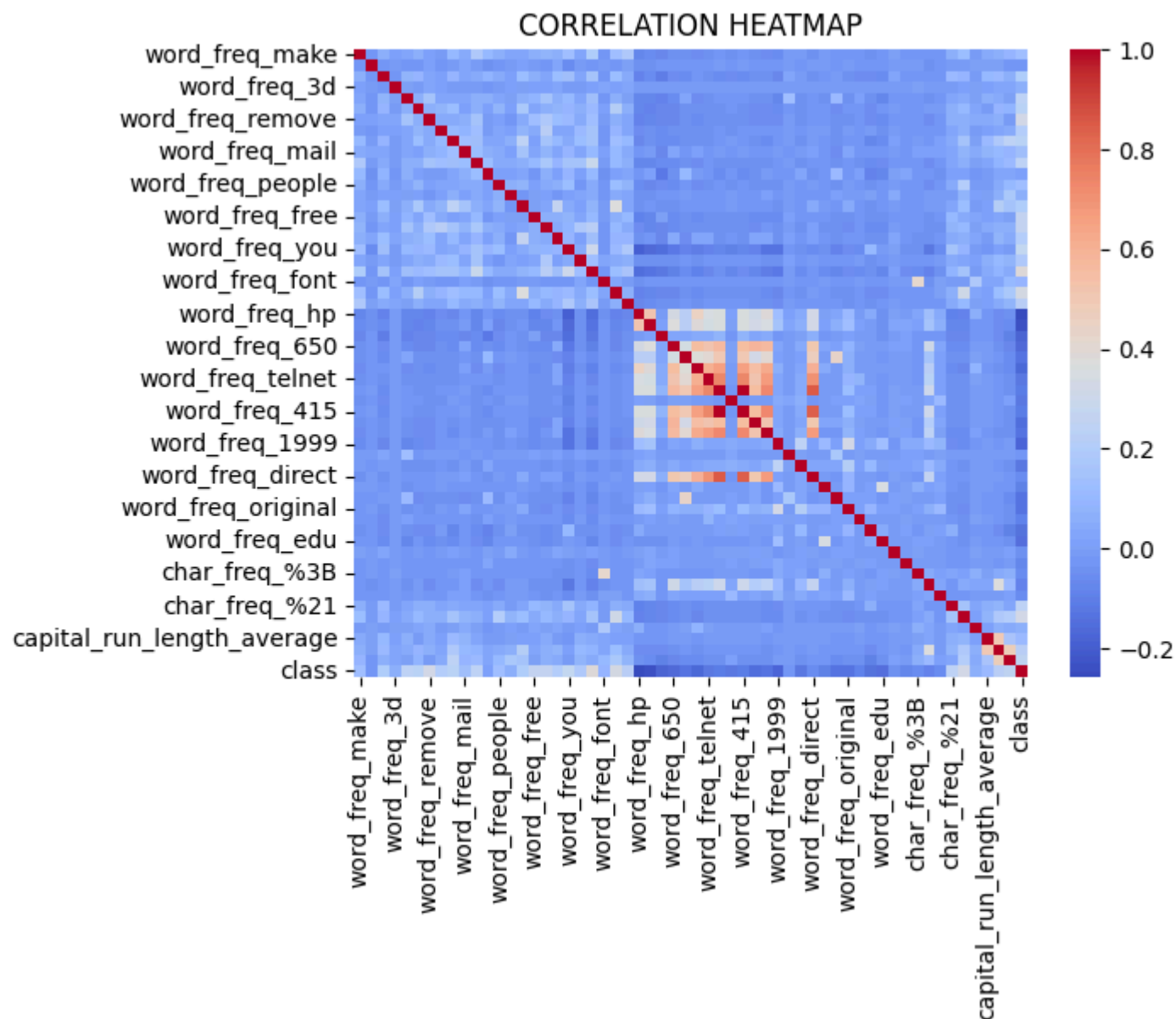
In [33]: #Normalizing or standardizing the features
from sklearn.preprocessing import StandardScaler
scaler=StandardScaler()
x_scaled=scaler.fit_transform(x)

In [34]: import matplotlib.pyplot as plt
import seaborn as sns
plt.hist(y,bins=30)
plt.title("HISTOGRAM :LOAN AMOUNT DISTRIBUTION")
plt.xlabel("Loan Amount")
plt.ylabel("Frequency")
plt.show()

```



```
In [35]: sns.heatmap(df.corr(), cmap="coolwarm")  
plt.title("CORRELATION HEATMAP")  
plt.show()
```



```
In [36]: from sklearn.decomposition import PCA
pca=PCA(n_components=0.95)
x_train_pca=pca.fit_transform(x_train)
```

```
x_test_pca=pca.transform(x_test)

n_components = pca.n_components_
explained_variance = pca.explained_variance_ratio_.sum() * 100

print("Chosen Components:", n_components)
print("Explained Variance:", explained_variance)
print("ORIGINAL SHAPE:", x_train.shape)
print("REDUCED SHAPE:", x_train_pca.shape)
```

Chosen Components: 45
Explained Variance: 95.00850406937306
ORIGINAL SHAPE: (3680, 54)
REDUCED SHAPE: (3680, 45)

```

/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:604: RuntimeWarning: divide by
zero encountered in matmul
    C = X.T @ X
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:604: RuntimeWarning: overflow
encountered in matmul
    C = X.T @ X
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:604: RuntimeWarning: invalid v
alue encountered in matmul
    C = X.T @ X
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: divide b
y zero encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: overflow
encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: invalid
value encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: divide b
y zero encountered in matmul
    X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: overflow
encountered in matmul
    X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: invalid
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: overflow

```

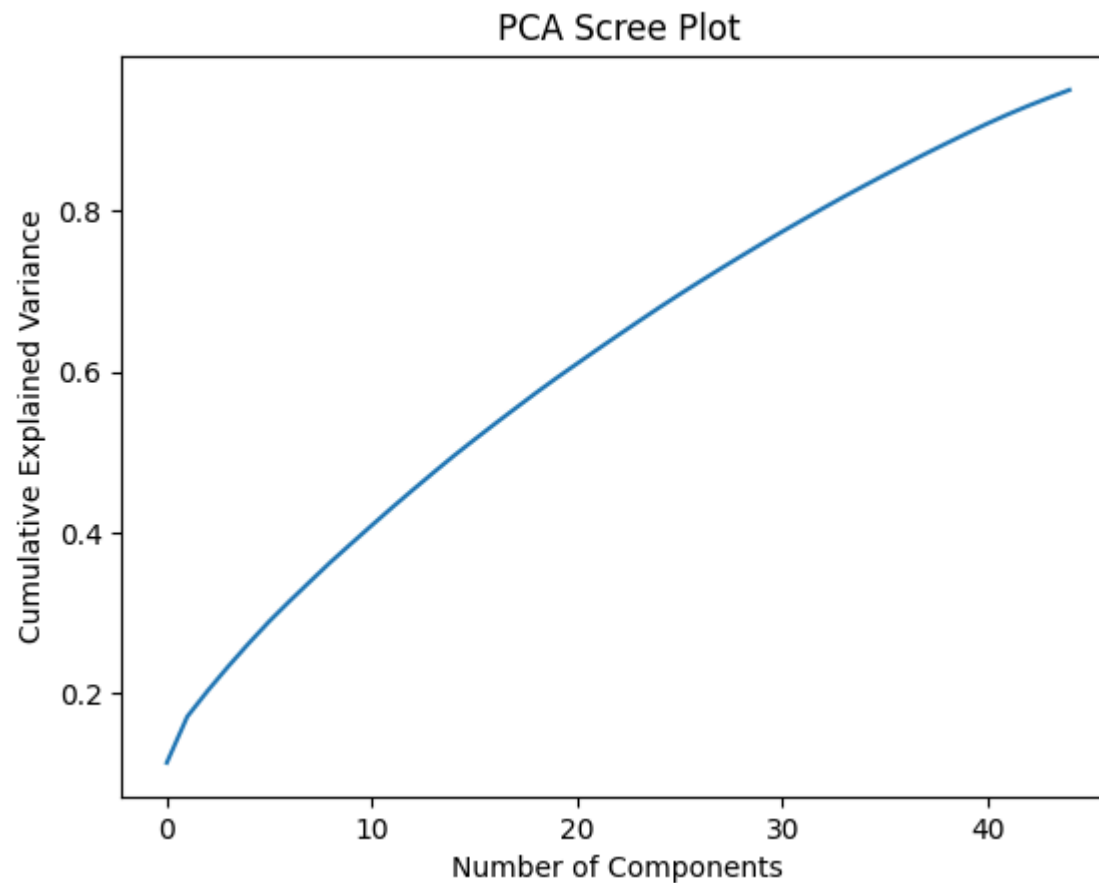
encountered in matmul

```
X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
```

/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:155: RuntimeWarning: invalid value encountered in matmul

```
X_transformed -= xp.reshape(self.mean_, (1, -1)) @ self.components_.T
```

```
In [37]: #Scree plot
plt.plot(np.cumsum(pca.explained_variance_ratio_))
plt.xlabel("Number of Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("PCA Scree Plot")
plt.show()
```



```
In [38]: pca_table=pd.DataFrame({"Settings":["With-PCA"],"Chosen Components":[f"{n_components}"],"Explained Variance (%)":[f
```

```
Out[38]:
```

	Settings	Chosen Components	Explained Variance (%)	Justification
0	With-PCA	45	95.00850406937306	Dimensionality reduced while retaining most da...

```
In [39]: from sklearn.linear_model import LogisticRegression
lr_params = {"C": [0.01, 0.1, 1, 10], "penalty": ["l2"], "solver": ["lbfgs"]}
#No PCA
lr_grid_no = GridSearchCV(LogisticRegression(max_iter=5000), lr_params, cv=5, scoring="accuracy")
lr_grid_no.fit(x_train, y_train)
best_lr_no = lr_grid_no.best_estimator_
y_pred_lr_no = best_lr_no.predict(x_test)
acc_lr_no = accuracy_score(y_test, y_pred_lr_no)
f1_lr_no = f1_score(y_test, y_pred_lr_no, average="weighted")
cv_lr_no = cross_val_score(best_lr_no, x_train, y_train, cv=5)

#With PCA
lr_grid_pca = GridSearchCV(LogisticRegression(max_iter=5000), lr_params, cv=5, scoring="accuracy")
lr_grid_pca.fit(x_train_pca, y_train)
best_lr_pca = lr_grid_pca.best_estimator_
y_pred_lr_pca = best_lr_pca.predict(x_test_pca)
acc_lr_pca = accuracy_score(y_test, y_pred_lr_pca)
f1_lr_pca = f1_score(y_test, y_pred_lr_pca, average="weighted")
cv_lr_pca = cross_val_score(best_lr_pca, x_train_pca, y_train, cv=5)
```

```

/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
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    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:330: RuntimeWarning: di
vide by zero encountered in matmul
    grad[:n_features] = X.T @ grad_pointwise + l2_reg_strength * weights
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:330: RuntimeWarning: ov
erflow encountered in matmul
    grad[:n_features] = X.T @ grad_pointwise + l2_reg_strength * weights

```



```

/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:330: RuntimeWarning: in
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: invalid value
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:330: RuntimeWarning: ov
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:330: RuntimeWarning: in
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: invalid value
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    ret = a @ b
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul

```



```

raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: overflow encountered in matmul
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:330: RuntimeWarning: overflow encountered in matmul
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:330: RuntimeWarning: in

```

```

valid value encountered in matmul
  grad[:n_features] = X.T @ grad_pointwise + l2_reg_strength * weights
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: divide by zero
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  ret = a @ b
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: invalid value
encountered in matmul
  ret = a @ b

```

```
In [40]: lr_table = pd.DataFrame({"Regularization (C) Tried": ["0.01, 0.1, 1, 10"], "Penalty": ["L2"], "Solver": ["lbfgs"], "Pe
lr_table
```

```
Out[40]:
```

	Regularization (C) Tried	Penalty	Solver	Performance (No PCA)	Performance (With PCA)
0	0.01, 0.1, 1, 10	L2	lbfgs	0.910966	0.914224

```
In [41]: lr_summary_table = pd.DataFrame({"Accuracy": [acc_lr_no, acc_lr_pca], "F1-Score": [f1_lr_no, f1_lr_pca]}, index=["Lo
lr_summary_table.round(6)
```

```
Out[41]:
```

	Accuracy	F1-Score
Logistic Regression (No PCA)	0.910966	0.910537
Logistic Regression (With PCA)	0.914224	0.913622

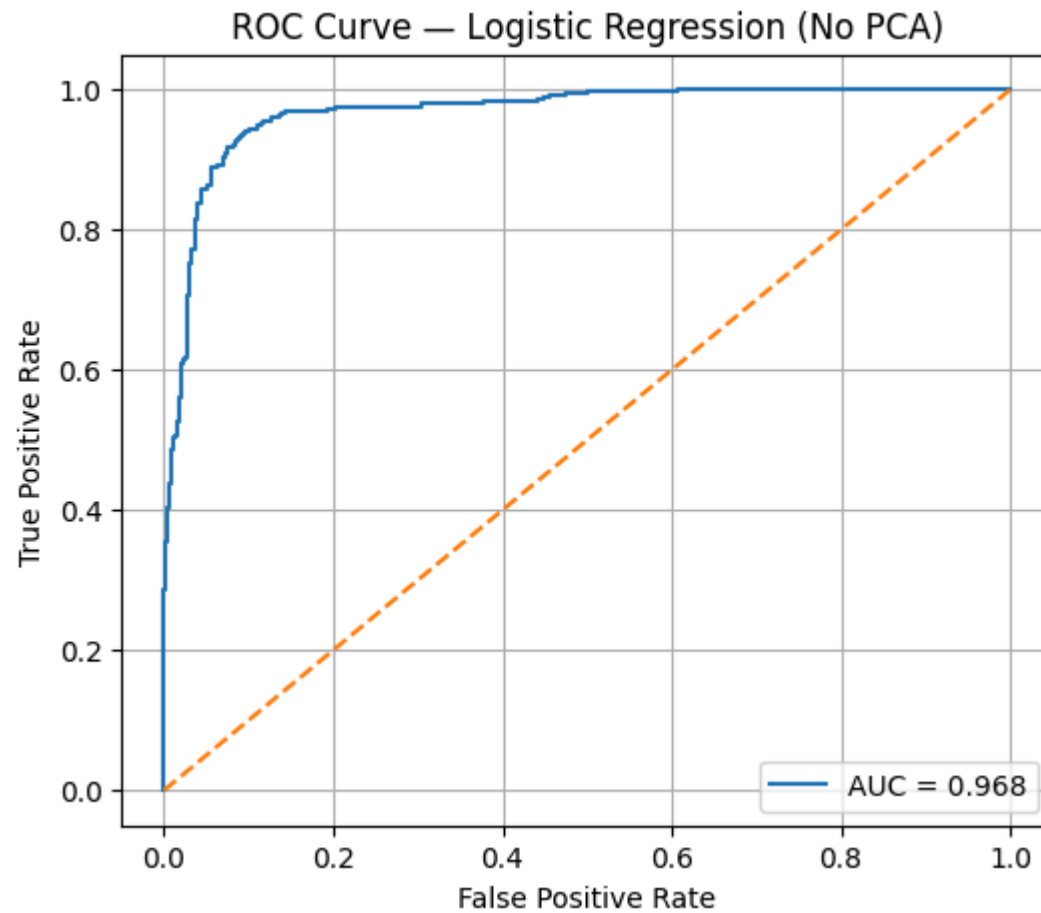
```
In [42]: # without pca
y_prob_lr_no = best_lr_no.predict_proba(x_test)[: , 1]
# ROC values
fpr, tpr, _ = roc_curve(y_test, y_prob_lr_no)
roc_auc = auc(fpr, tpr)
# ROC Plot
plt.figure(figsize=(6,5))
plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
plt.plot([0,1], [0,1], '--')
plt.title("ROC Curve - Logistic Regression (No PCA)")
plt.xlabel("False Positive Rate")

```

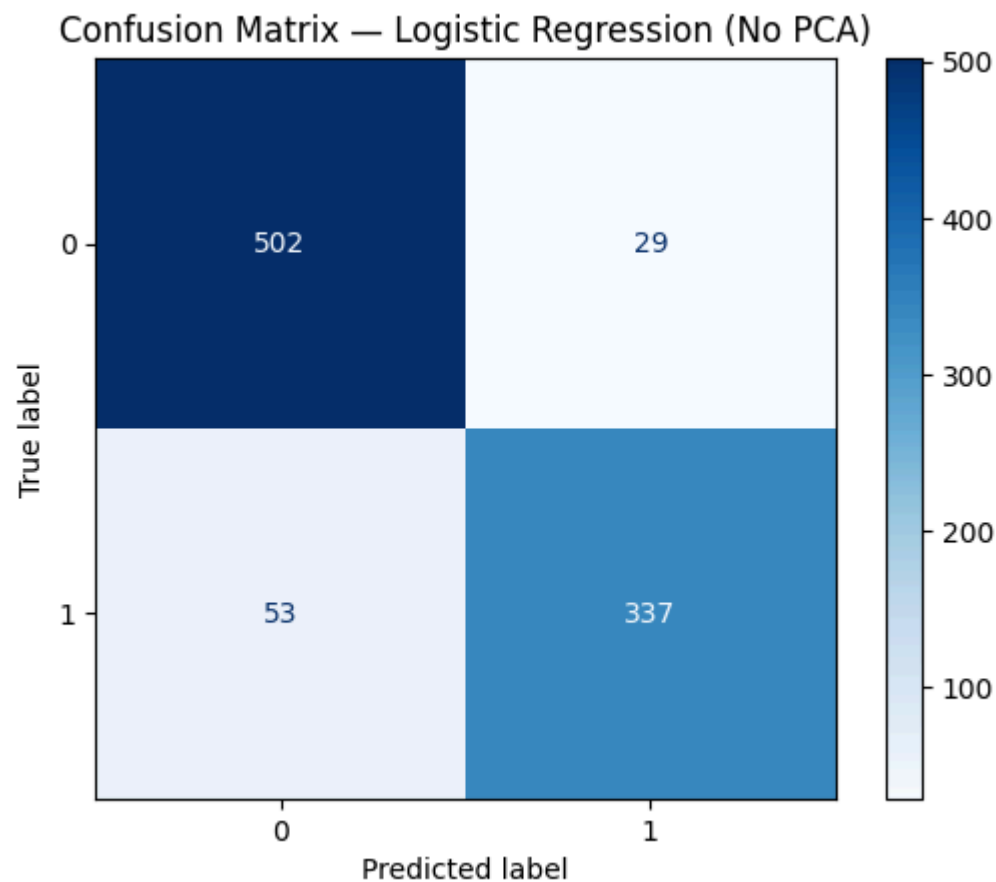
```
plt.ylabel("True Positive Rate")
plt.legend()
plt.grid(True)
plt.show()

# Confusion Matrix
y_pred_lr_no = best_lr_no.predict(x_test)
cm = confusion_matrix(y_test, y_pred_lr_no)
ConfusionMatrixDisplay(cm).plot(cmap="Blues")
plt.title("Confusion Matrix – Logistic Regression (No PCA)")
plt.show()
```

```
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: divide by zero
encountered in matmul
    ret = a @ b
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: overflow encou
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: invalid value
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```



```
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: invalid value encountered in matmul
  ret = a @ b
```



```
In [43]: # with pca
y_prob_lr_pca = best_lr_pca.predict_proba(x_test_pca)[: , 1]

# ROC values
fpr, tpr, _ = roc_curve(y_test, y_prob_lr_pca)
roc_auc = auc(fpr, tpr)

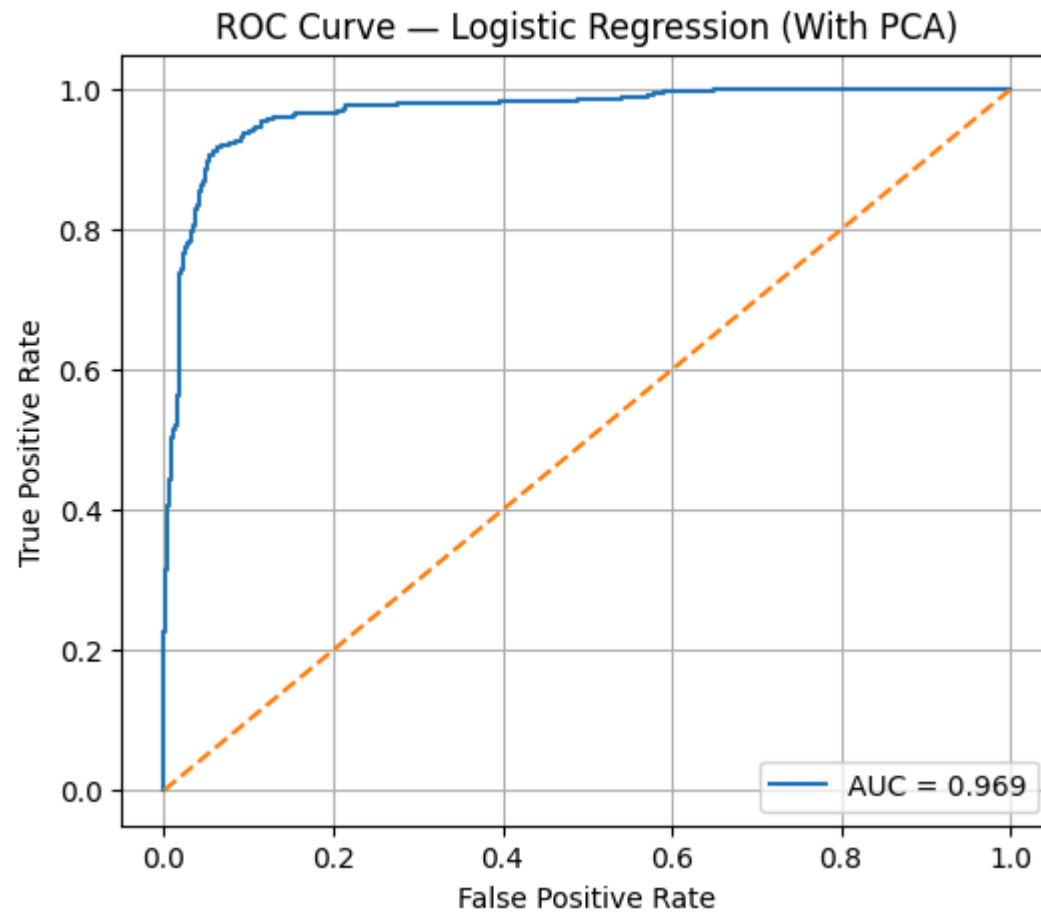
# ROC Plot
plt.figure(figsize=(6,5))
plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
plt.plot([0,1], [0,1], '--')
plt.title("ROC Curve – Logistic Regression (With PCA)")
plt.xlabel("False Positive Rate")
```

```
plt.ylabel("True Positive Rate")
plt.legend()
plt.grid(True)
plt.show()

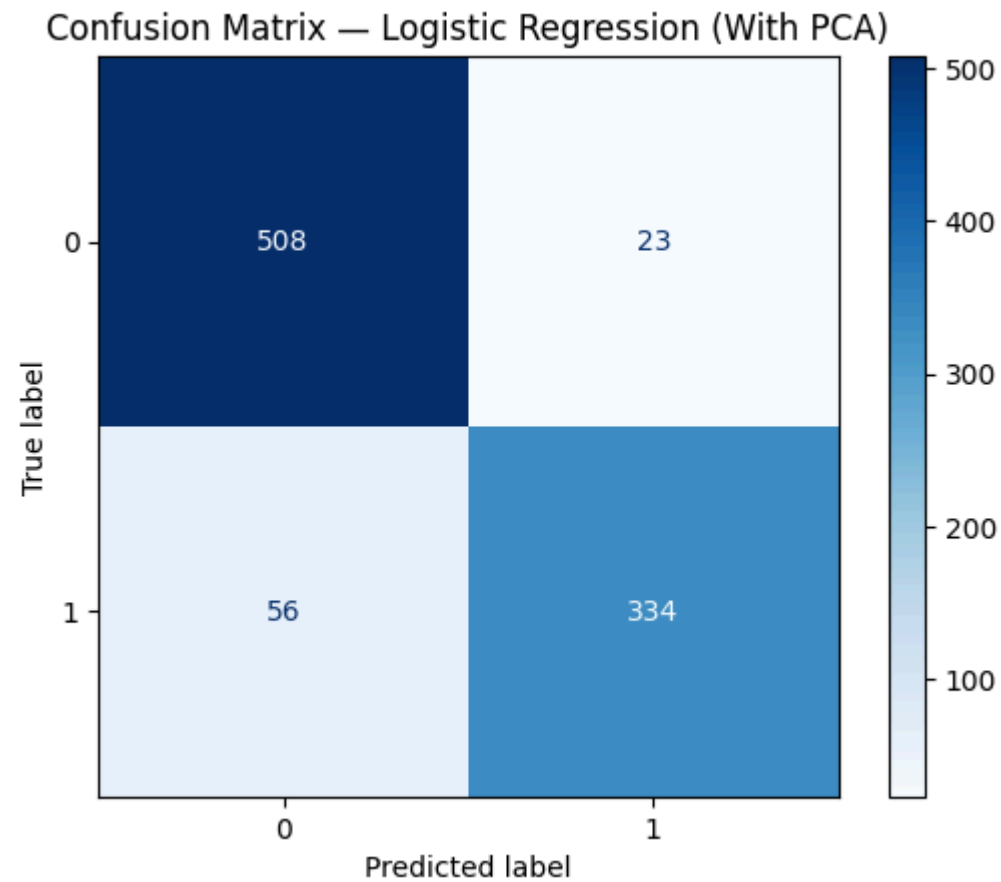
# Confusion Matrix
y_pred_lr_pca = best_lr_pca.predict(x_test_pca)
cm = confusion_matrix(y_test, y_pred_lr_pca)

ConfusionMatrixDisplay(cm).plot(cmap="Blues")
plt.title("Confusion Matrix – Logistic Regression (With PCA)")
plt.show()
```

```
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: divide by zero
encountered in matmul
  ret = a @ b
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: overflow encou
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  ret = a @ b
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```



```
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: divide by zero encountered in matmul
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/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/utils/extmath.py:203: RuntimeWarning: invalid value encountered in matmul
  ret = a @ b
```



Decision Tree, Random Forest, AdaBoost, Gradient Boosting, XGBoost, Stacking

Dataset used

- Breast Cancer dataset
- From assignment-4

```
In [44]: df_all=pd.read_csv("wdbc.data")
print("Number of Samples :", df_all.shape[0])
print("Number of Features :", df_all.shape[1] - 1)
```


Number of Samples : 568

Number of Features : 31

```
In [45]: #STATISTICS
df_all.describe()
```

```
Out [45]:
```

	842302	17.99	10.38	122.8	1001	0.1184	0.2776	0.3001	0.1471	
count	5.680000e+02	568.000000	568.000000	568.000000	568.000000	568.000000	568.000000	568.000000	568.000000	568.
mean	3.042382e+07	14.120491	19.305335	91.914754	654.279754	0.096321	0.104036	0.088427	0.048746	0
std	1.251246e+08	3.523416	4.288506	24.285848	351.923751	0.014046	0.052355	0.079294	0.038617	0
min	8.670000e+03	6.981000	9.710000	43.790000	143.500000	0.052630	0.019380	0.000000	0.000000	0.
25%	8.692225e+05	11.697500	16.177500	75.135000	420.175000	0.086290	0.064815	0.029540	0.020310	0
50%	9.061570e+05	13.355000	18.855000	86.210000	548.750000	0.095865	0.092525	0.061400	0.033455	0
75%	8.825022e+06	15.780000	21.802500	103.875000	782.625000	0.105300	0.130400	0.129650	0.073730	0.
max	9.113205e+08	28.110000	39.280000	188.500000	2501.000000	0.163400	0.345400	0.426800	0.201200	0.

8 rows x 31 columns

```
In [46]: df_all.columns
```

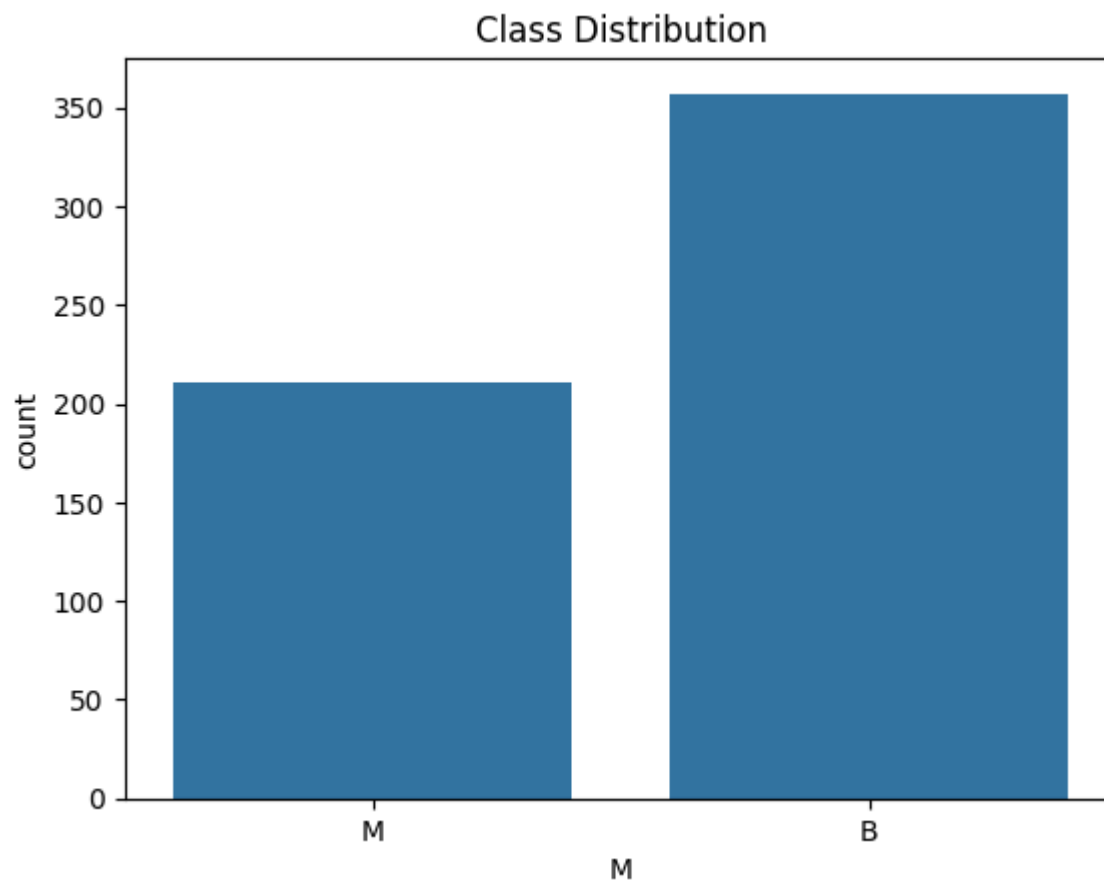
```
Out [46]: Index(['842302', 'M', '17.99', '10.38', '122.8', '1001', '0.1184', '0.2776',
                '0.3001', '0.1471', '0.2419', '0.07871', '1.095', '0.9053', '8.589',
                '153.4', '0.006399', '0.04904', '0.05373', '0.01587', '0.03003',
                '0.006193', '25.38', '17.33', '184.6', '2019', '0.1622', '0.6656',
                '0.7119', '0.2654', '0.4601', '0.1189'],
                dtype='object')
```

```
In [47]: x=df_all.drop("M",axis=1)
y=df_all["M"]
```

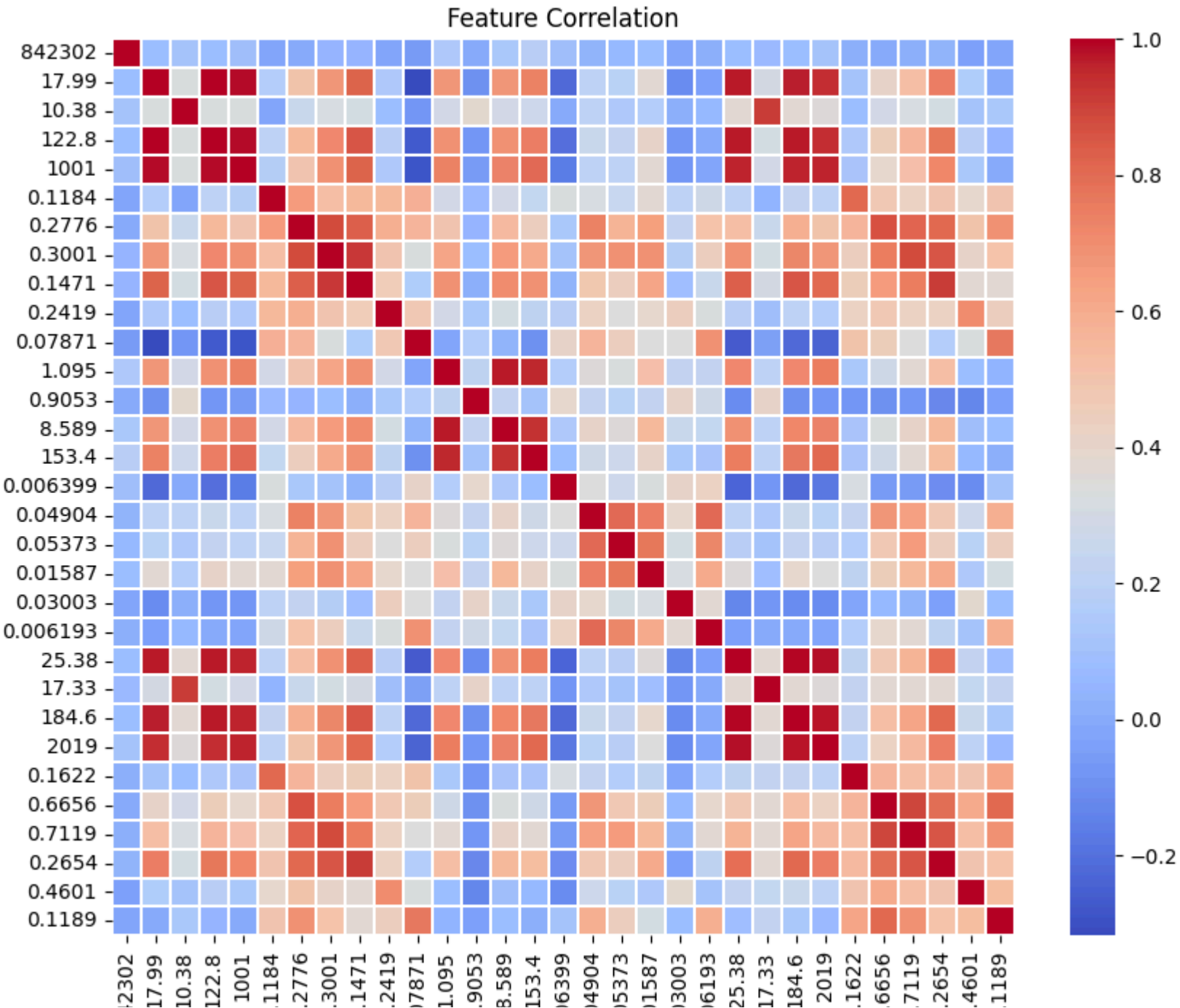
```
In [48]: from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
y = le.fit_transform(y)
```

```
In [49]: #Normalizing or standardizing the features  
from sklearn.preprocessing import StandardScaler  
scaler=StandardScaler()  
x_scaled=scaler.fit_transform(x,y)
```

```
In [50]: import matplotlib.pyplot as plt  
import seaborn as sns  
sns.countplot(x="M",data=df_all)  
plt.title("Class Distribution")  
plt.show()
```



```
In [51]: plt.figure(figsize=(10,8))
correlation_matrix=df_all.drop(columns='M').corr()
sns.heatmap(correlation_matrix,cmap="coolwarm",linewidths=0.2,cbar=True)
plt.title("Feature Correlation")
plt.show()
```



84 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

```
In [52]: #splitting data
from sklearn.model_selection import train_test_split, GridSearchCV, cross_val_score
x_train, x_test, y_train, y_test = train_test_split(x_scaled, y, test_size=0.2, random_state=42)
```

```
In [53]: pca = PCA(n_components=0.95)
x_train_pca = pca.fit_transform(x_train)
x_test_pca = pca.transform(x_test)

n_components = pca.n_components_
explained_variance = pca.explained_variance_ratio_.sum() * 100

print("Chosen Components:", n_components)
print("Explained Variance:", explained_variance)
print("ORIGINAL SHAPE:", x_train.shape)
print("REDUCED SHAPE:", x_train_pca.shape)
```

Chosen Components: 11
Explained Variance: 95.34461023636725
ORIGINAL SHAPE: (454, 31)
REDUCED SHAPE: (454, 11)

```

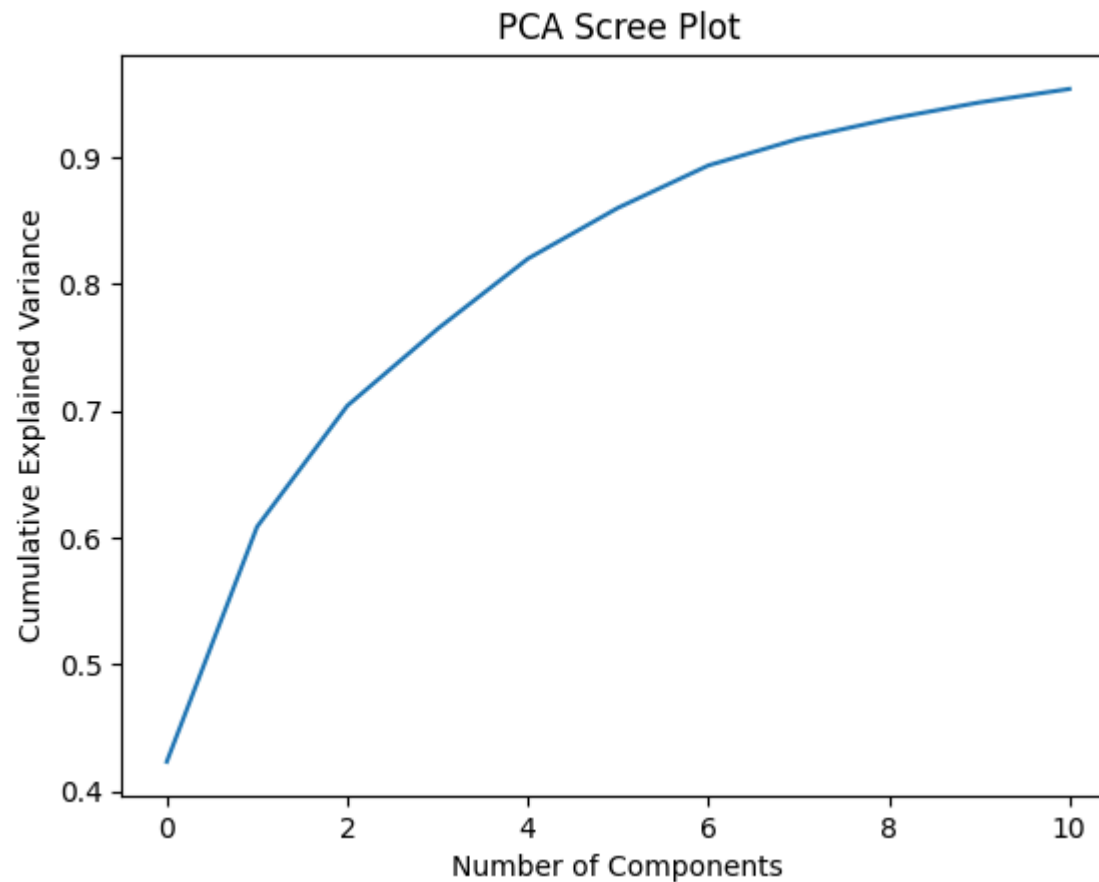
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:604: RuntimeWarning: divide by
zero encountered in matmul
    C = X.T @ X
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:604: RuntimeWarning: overflow
encountered in matmul
    C = X.T @ X
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_pca.py:604: RuntimeWarning: invalid v
alue encountered in matmul
    C = X.T @ X
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: divide b
y zero encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: overflow
encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: invalid
value encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: divide b
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    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: overflow
encountered in matmul
    X_transformed = X @ self.components_.T
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/decomposition/_base.py:148: RuntimeWarning: invalid
value encountered in matmul
    X_transformed = X @ self.components_.T

```

```

In [54]: #Scree plot
plt.plot(np.cumsum(pca.explained_variance_ratio_))
plt.xlabel("Number of Components")
plt.ylabel("Cumulative Explained Variance")
plt.title("PCA Scree Plot")
plt.show()

```



```
In [55]: pca_table=pd.DataFrame({"Settings":["With-PCA"],"Chosen Components":[f"{n_components}"],"Explained Variance (%)":[f
pca_table
```

```
Out[55]:
```

	Settings	Chosen Components	Explained Variance (%)	Justification
0	With-PCA	11	95.34461023636725	Dimensionality reduced while retaining most da...

```
In [57]: from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier, GradientBoostingClassifier, StackingClassi
from xgboost import XGBClassifier
dt_params = {"max_depth": [3,5,10],"min_samples_split": [2,5,10]}
rf_params = {"n_estimators": [50,100],"max_depth": [5,10]}
```

```

ada_params = {"n_estimators": [50,100],"learning_rate": [0.5,1]}
gb_params = {"n_estimators": [50,100],"learning_rate": [0.05,0.1]}
xgb_params = {"n_estimators": [50,100],"max_depth": [3,5],"learning_rate": [0.05,0.1]}

def evaluate_model(model, params, x_tr, x_te):
    grid = GridSearchCV(model, params, cv=5, scoring="accuracy")
    grid.fit(x_tr, y_train)
    best = grid.best_estimator_
    y_pred = best.predict(x_te)
    acc = accuracy_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred, average="weighted")
    cv = cross_val_score(best, x_tr, y_train, cv=5)
    return acc, f1, cv, best

```

```

In [58]: #without pca
dt_no = evaluate_model(DecisionTreeClassifier(), dt_params, x_train, x_test)
rf_no = evaluate_model(RandomForestClassifier(), rf_params, x_train, x_test)
ada_no = evaluate_model(AdaBoostClassifier(), ada_params, x_train, x_test)
gb_no = evaluate_model(GradientBoostingClassifier(), gb_params, x_train, x_test)
xgb_no = evaluate_model(XGBClassifier(eval_metric="logloss"), xgb_params, x_train, x_test)

```

```

In [59]: #with pca
dt_pca = evaluate_model(DecisionTreeClassifier(), dt_params, x_train_pca, x_test_pca)
rf_pca = evaluate_model(RandomForestClassifier(), rf_params, x_train_pca, x_test_pca)
ada_pca = evaluate_model(AdaBoostClassifier(), ada_params, x_train_pca, x_test_pca)
gb_pca = evaluate_model(GradientBoostingClassifier(), gb_params, x_train_pca, x_test_pca)
xgb_pca = evaluate_model(XGBClassifier(eval_metric="logloss"), xgb_params, x_train_pca, x_test_pca)

```

```

In [79]: # No PCA
stack_model.fit(x_train, y_train)
pred_no = stack_model.predict(x_test)
acc_stack_no = accuracy_score(y_test, pred_no)
f1_stack_no = f1_score(y_test, pred_no, average="weighted")

# With PCA
stack_model.fit(x_train_pca, y_train)
pred_pca = stack_model.predict(x_test_pca)
acc_stack_pca = accuracy_score(y_test, pred_pca)
f1_stack_pca = f1_score(y_test, pred_pca, average="weighted")

```



```

/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept

```

```

In [80]: def plot_roc_cm(model, X_test_used, y_test, title):
        y_prob = model.predict_proba(X_test_used)[:, 1]

        # ROC
        fpr, tpr, _ = roc_curve(y_test, y_prob)
        roc_auc = auc(fpr, tpr)

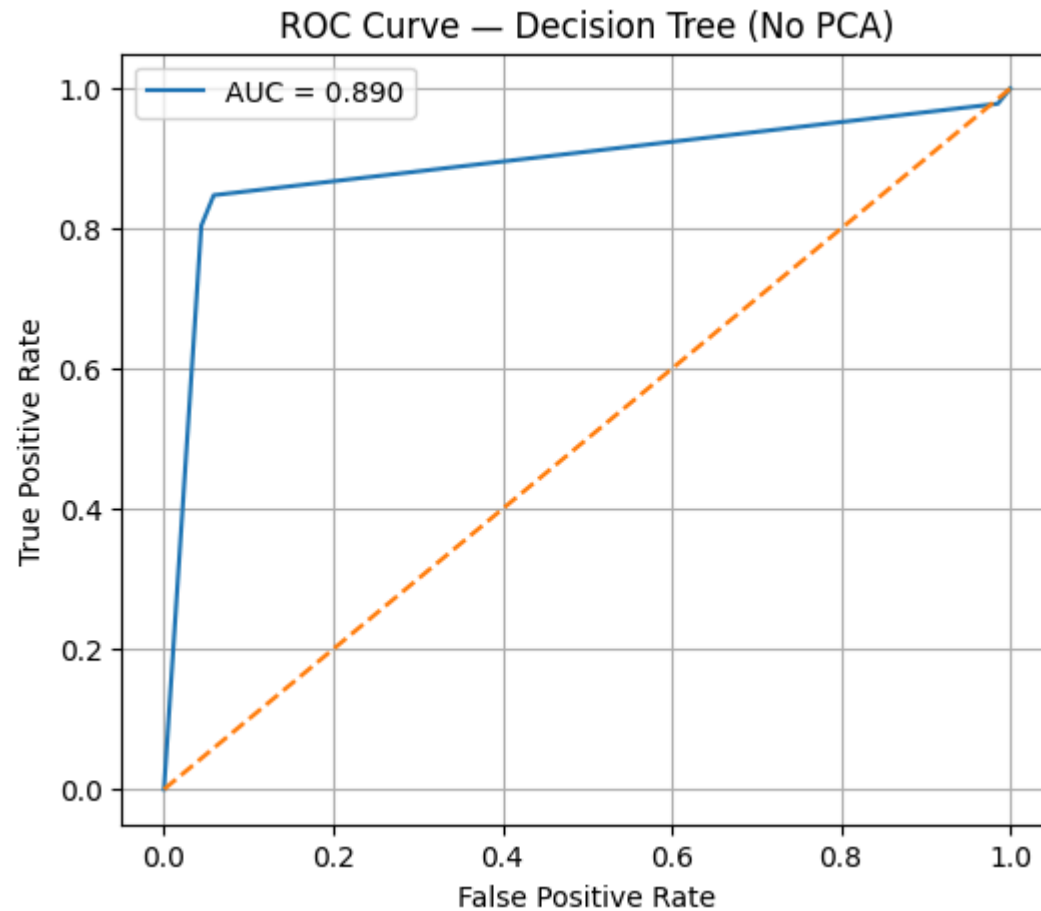
        plt.figure(figsize=(6,5))
        plt.plot(fpr, tpr, label=f"AUC = {roc_auc:.3f}")
        plt.plot([0,1],[0,1], '--')
        plt.title(f"ROC Curve - {title}")
        plt.xlabel("False Positive Rate")
        plt.ylabel("True Positive Rate")
        plt.legend()
        plt.grid(True)
        plt.show()

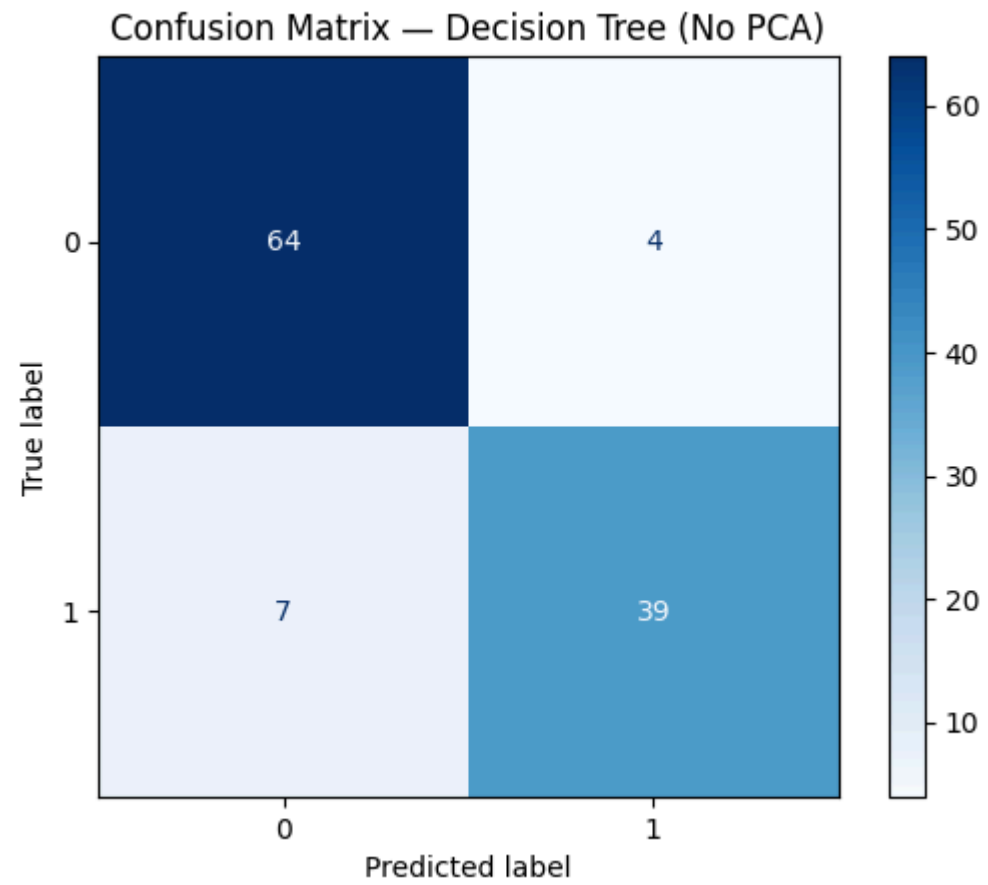
        # Confusion Matrix
        y_pred = model.predict(X_test_used)
        cm = confusion_matrix(y_test, y_pred)

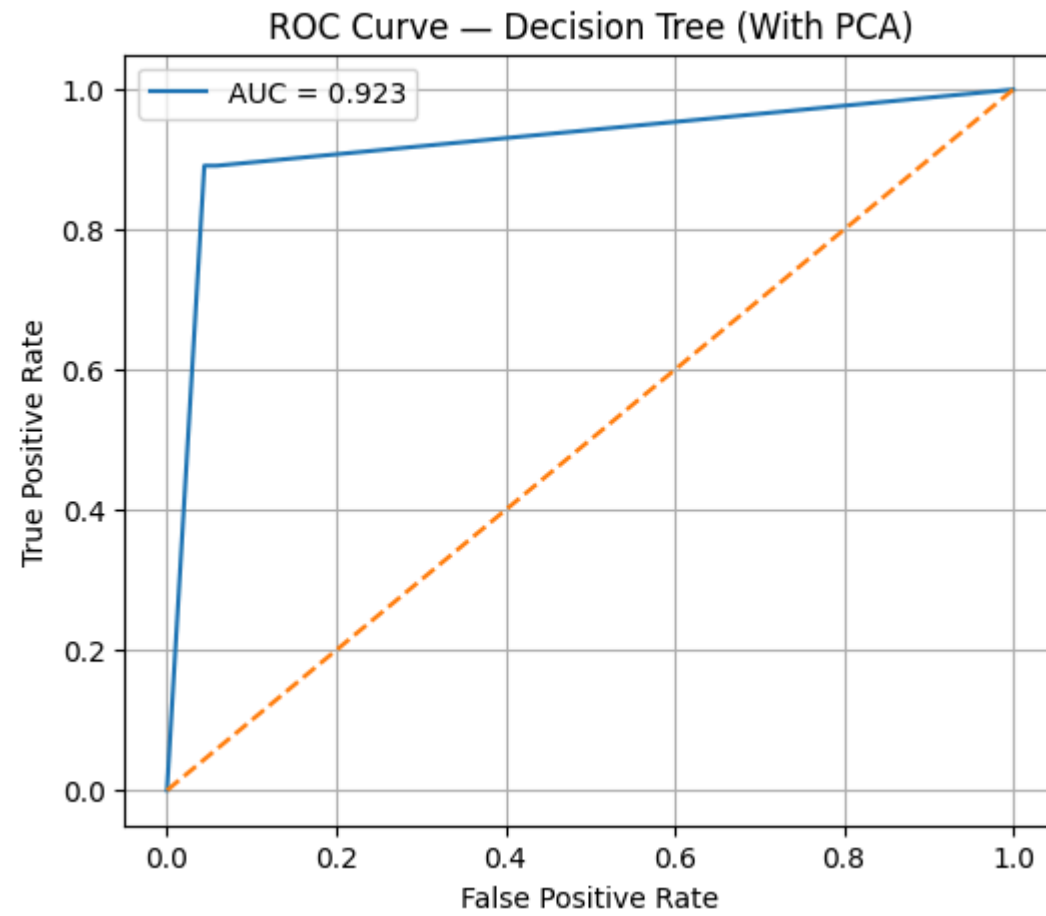
```

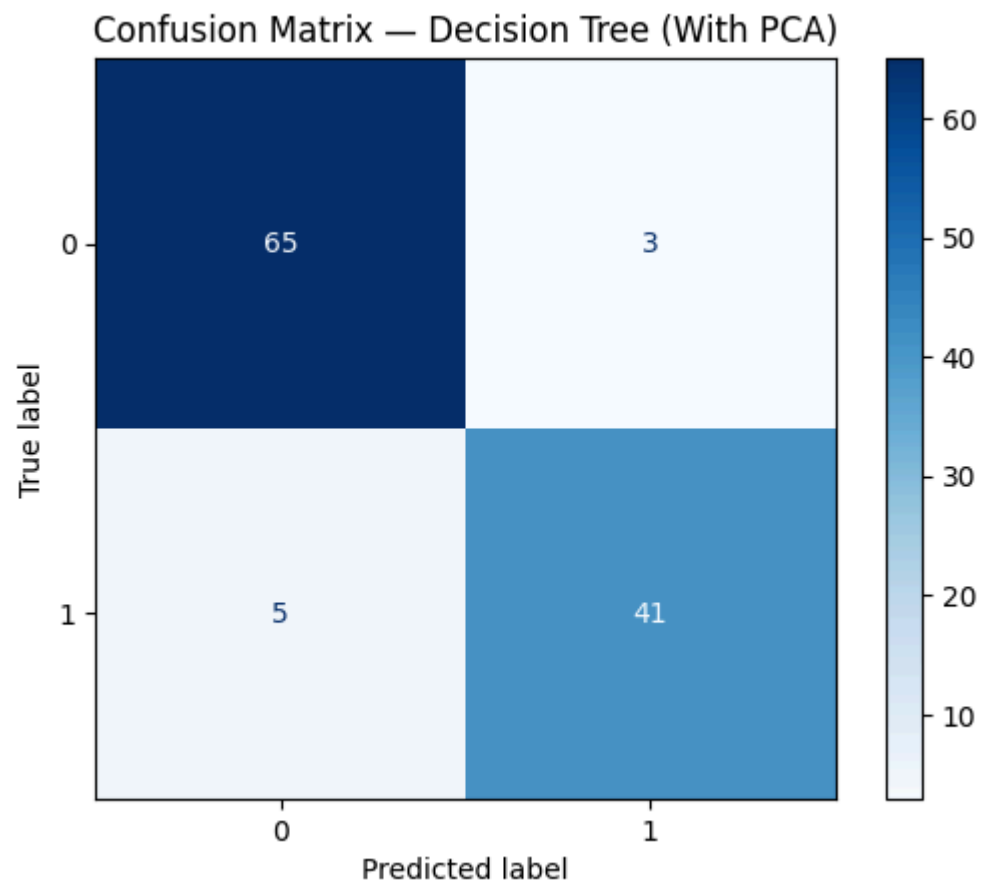
```
ConfusionMatrixDisplay(cm).plot(cmap="Blues")  
plt.title(f"Confusion Matrix - {title}")  
plt.show()
```

```
In [81]: plot_roc_cm(dt_no[3], x_test, y_test, "Decision Tree (No PCA)")  
plot_roc_cm(dt_pca[3], x_test_pca, y_test, "Decision Tree (With PCA)")
```

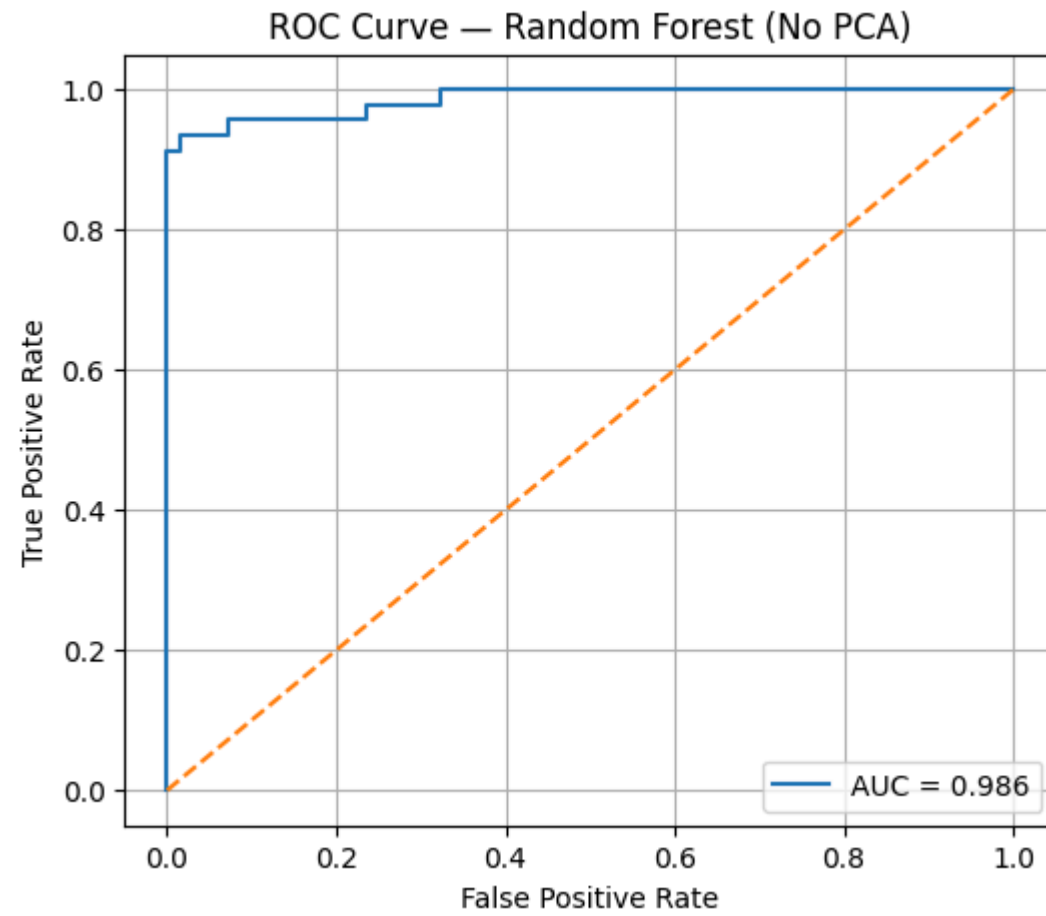


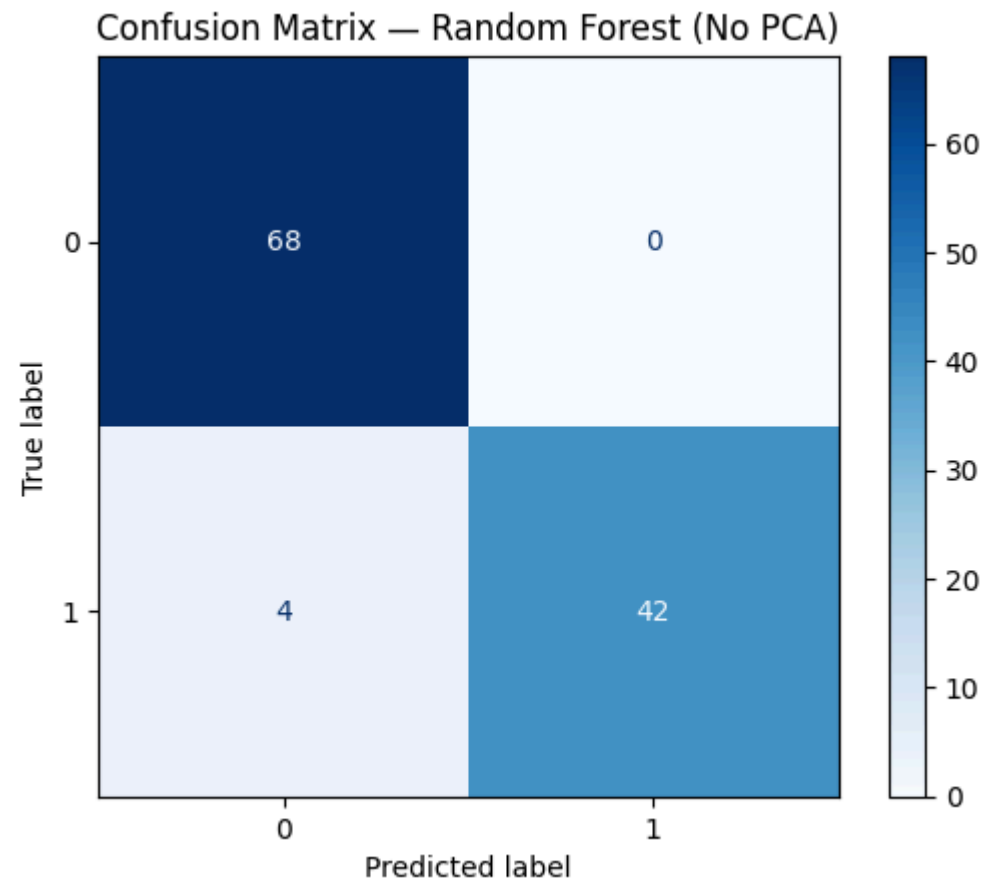


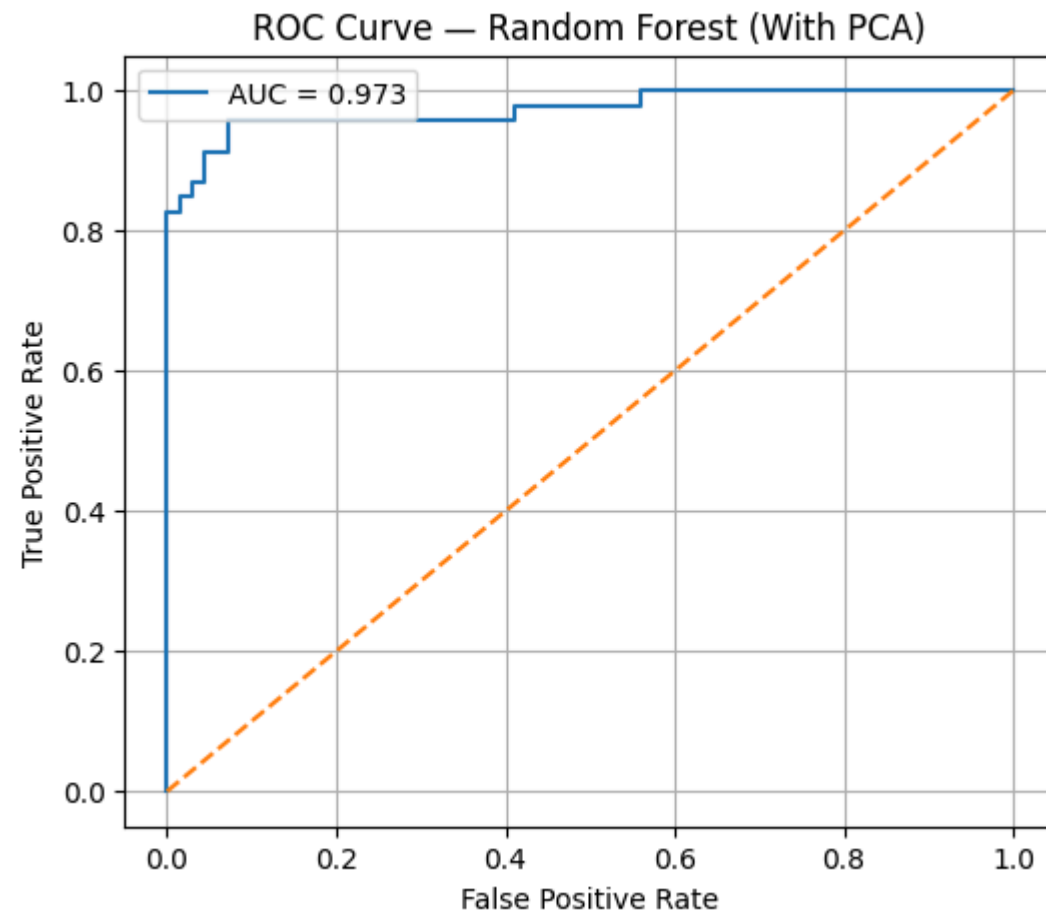


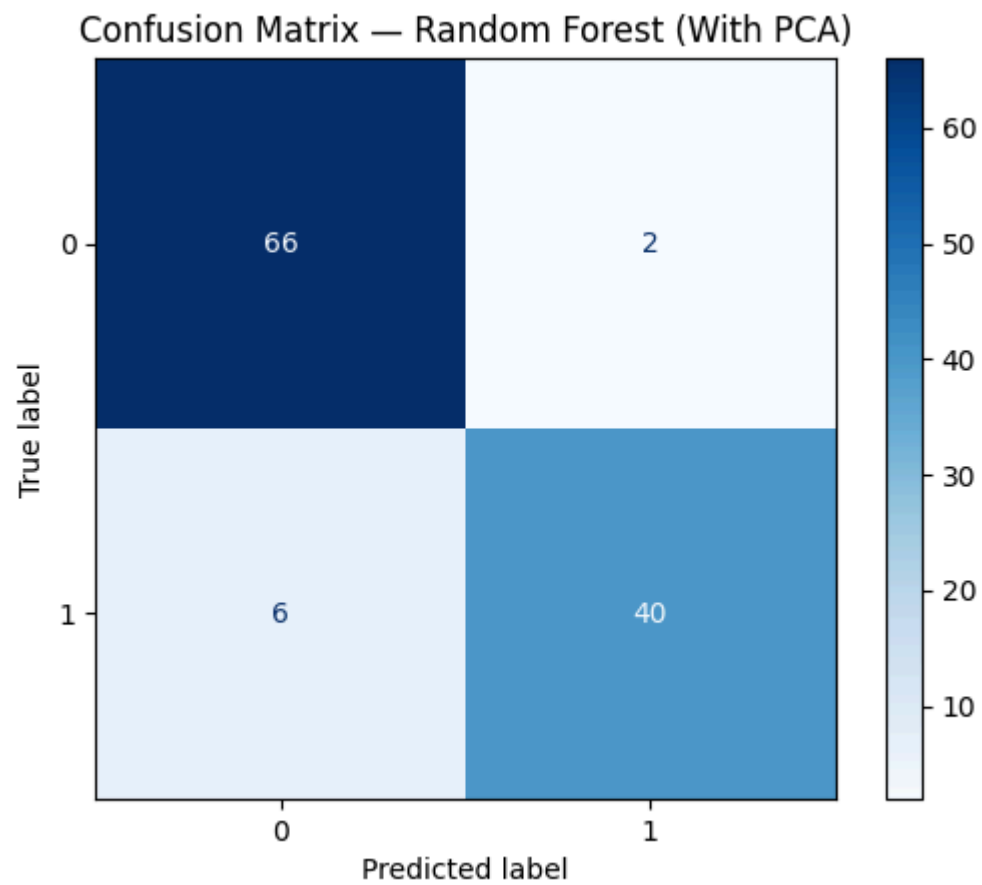


```
In [82]: plot_roc_cm(rf_no[3], x_test, y_test, "Random Forest (No PCA)")  
plot_roc_cm(rf_pca[3], x_test_pca, y_test, "Random Forest (With PCA)")
```

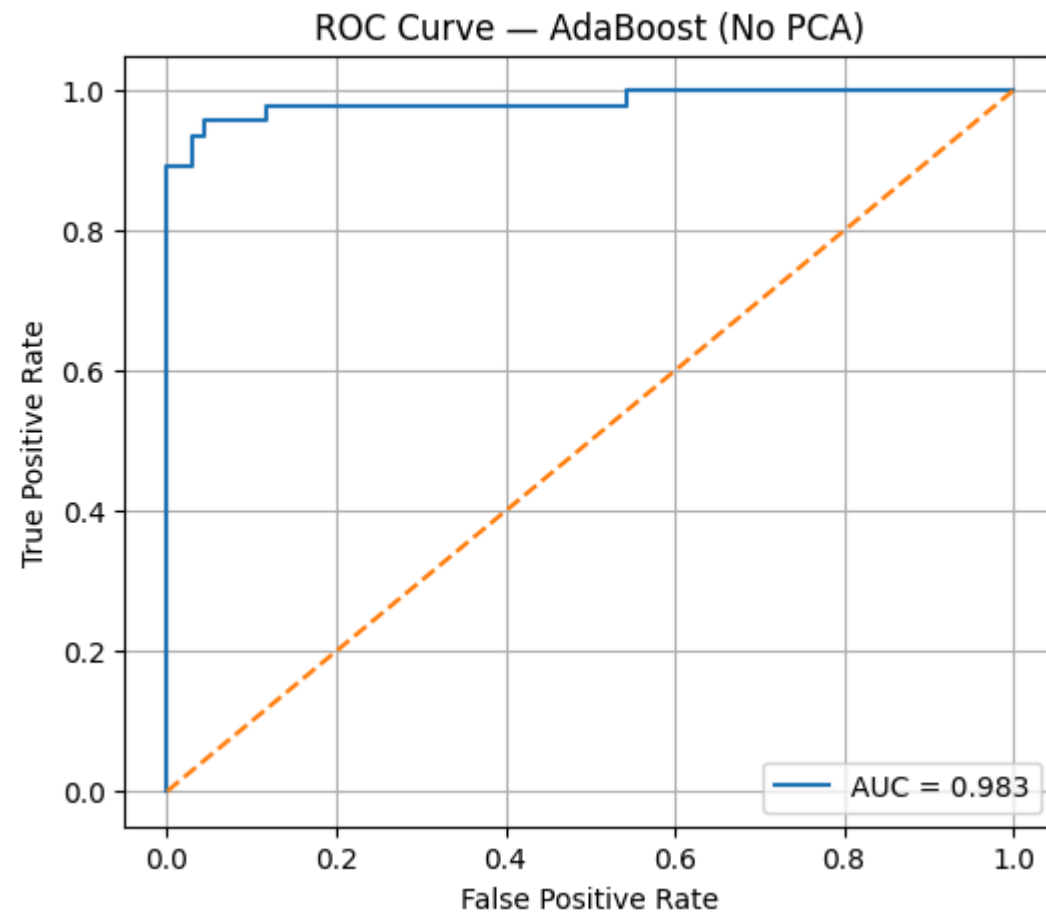


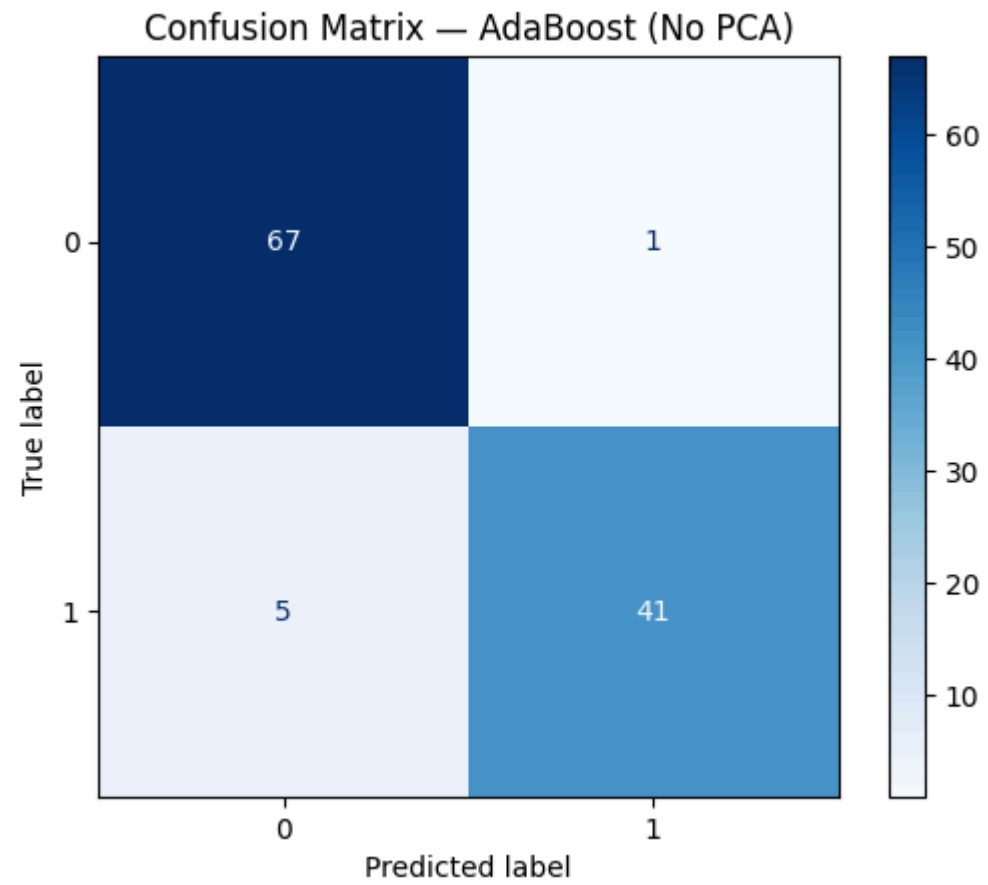


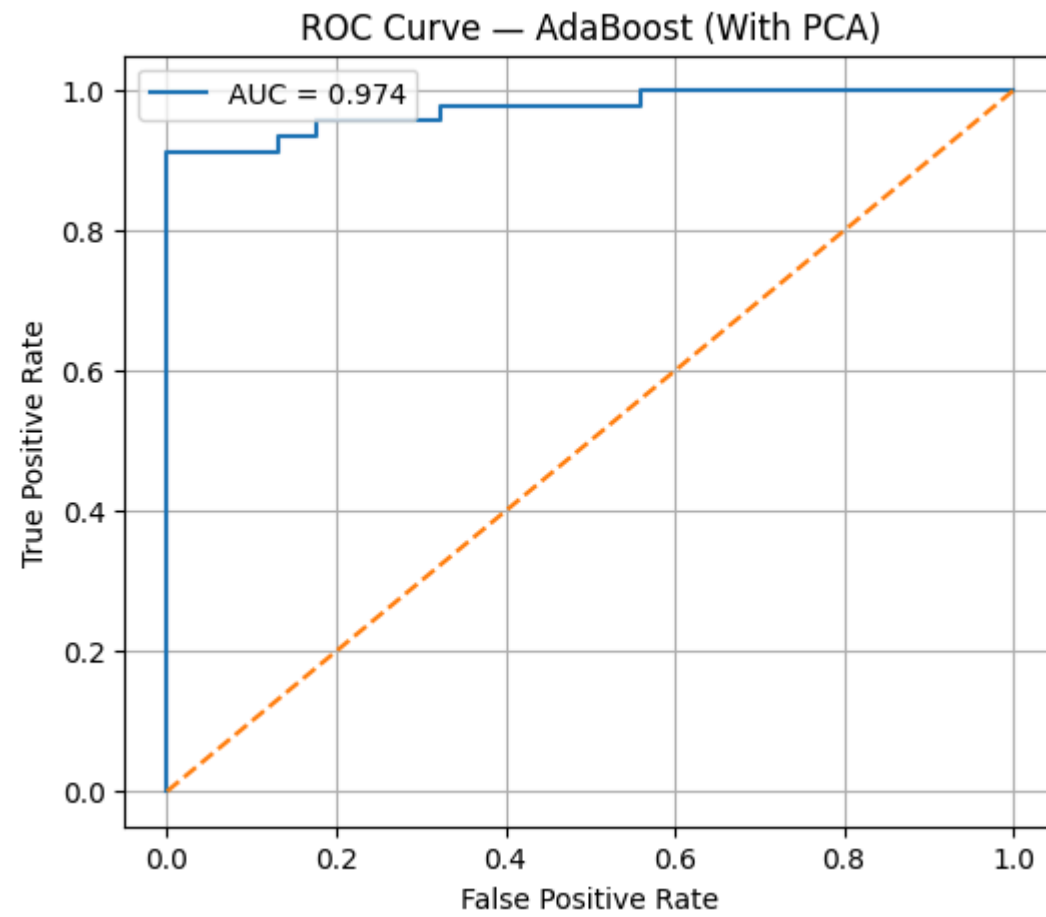


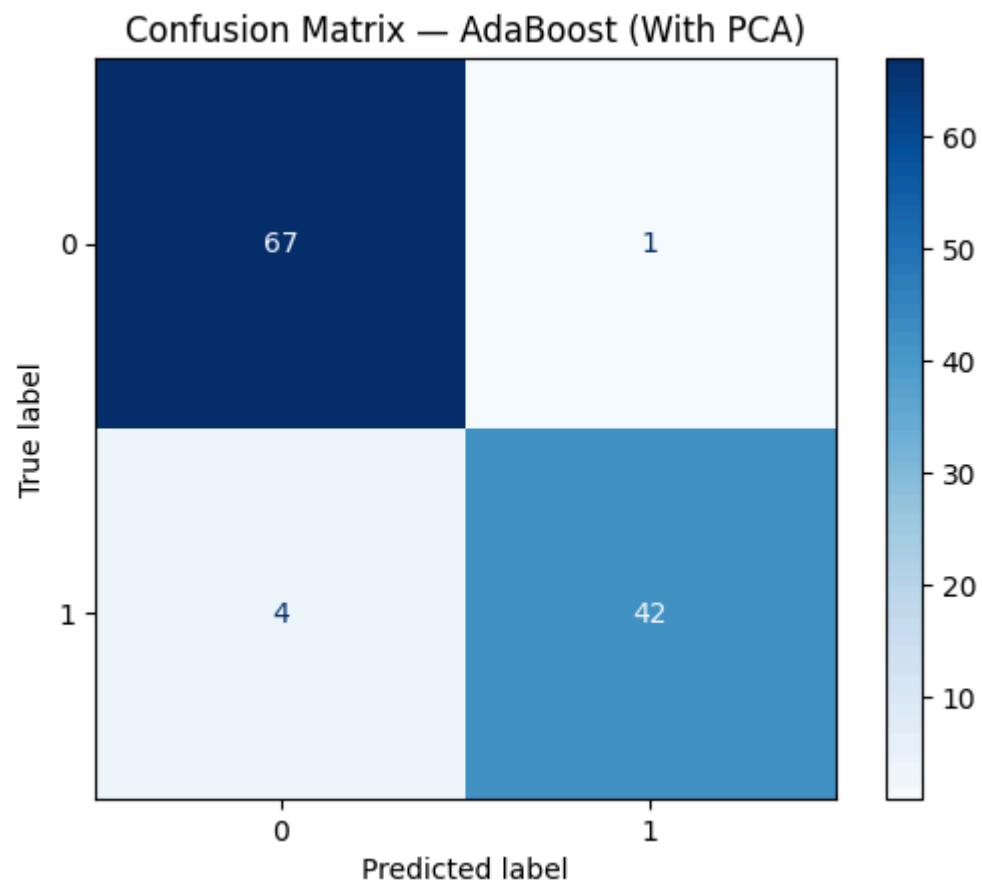


```
In [83]: plot_roc_cm(ada_no[3], x_test, y_test, "AdaBoost (No PCA)")  
plot_roc_cm(ada_pca[3], x_test_pca, y_test, "AdaBoost (With PCA)")
```

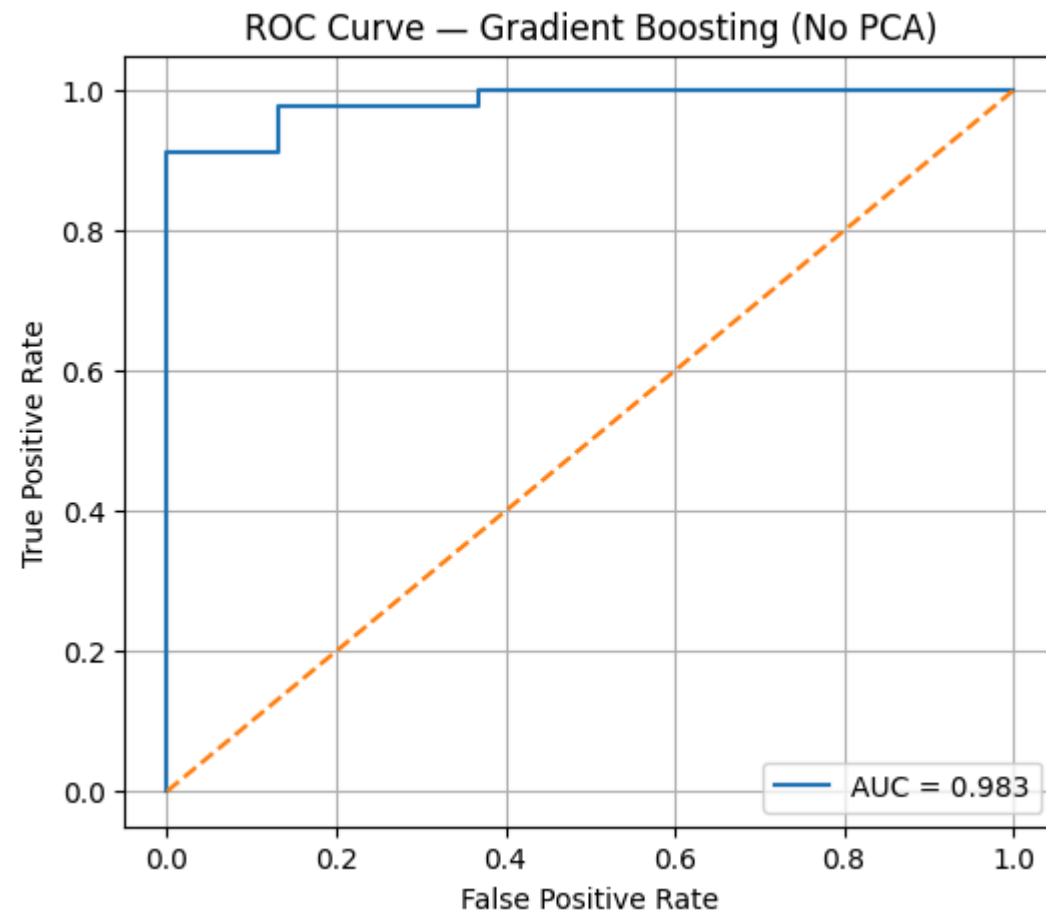


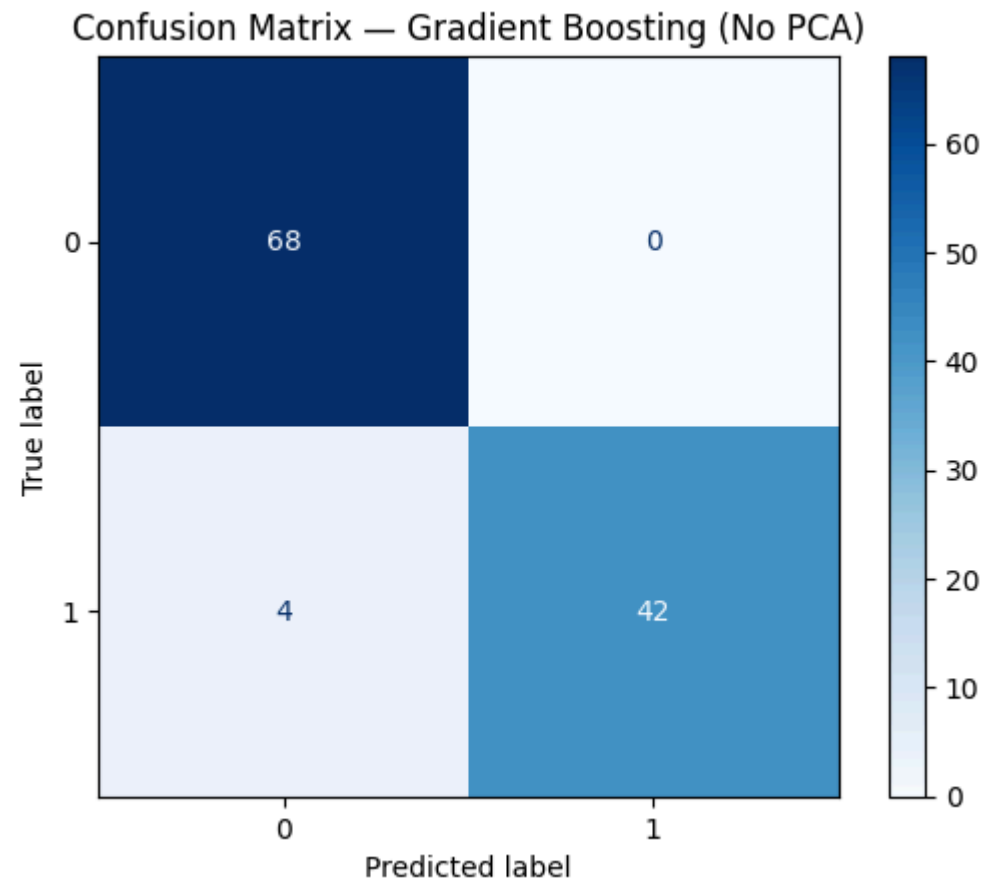


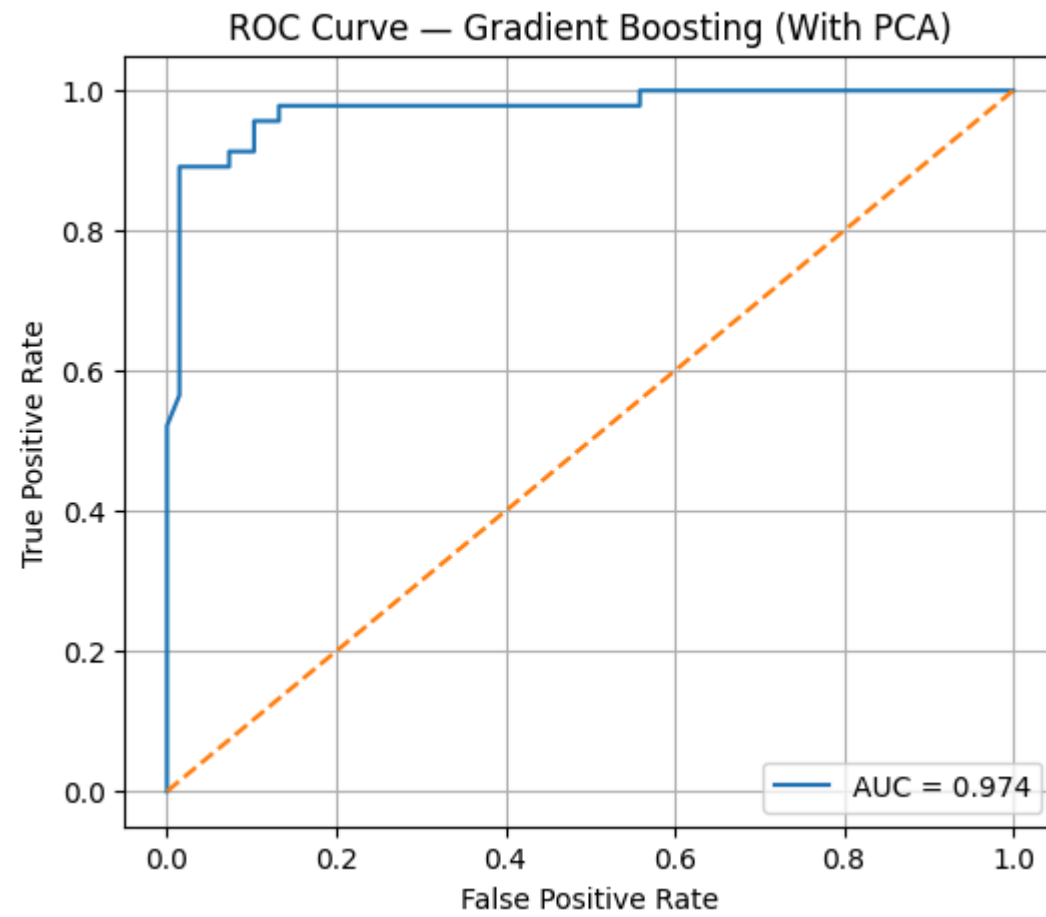


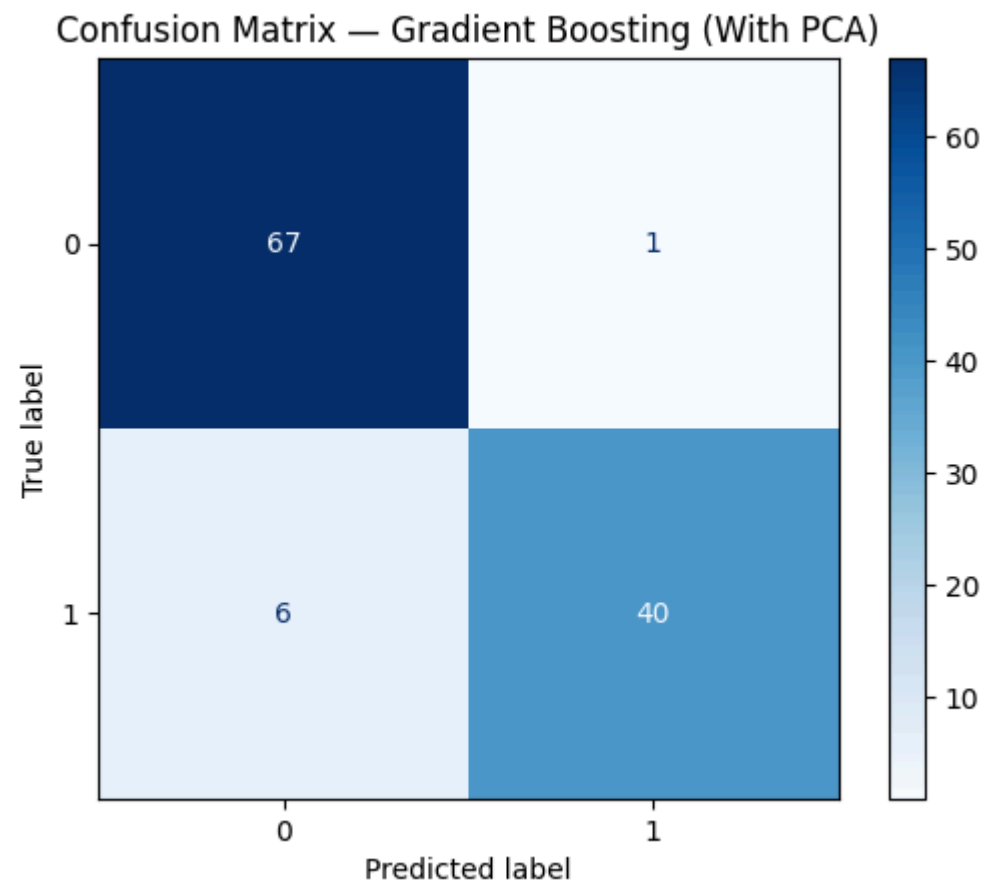


```
In [84]: plot_roc_cm(gb_no[3], x_test, y_test, "Gradient Boosting (No PCA)")  
plot_roc_cm(gb_pca[3], x_test_pca, y_test, "Gradient Boosting (With PCA)")
```

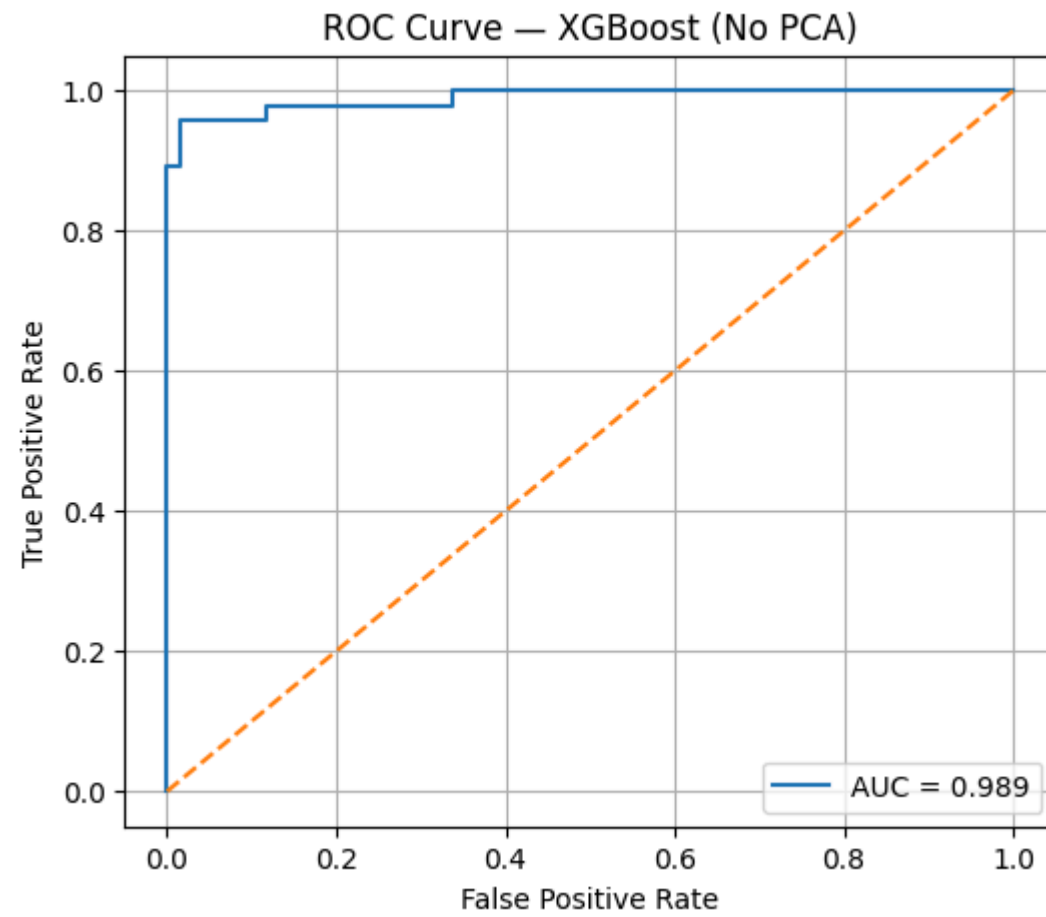


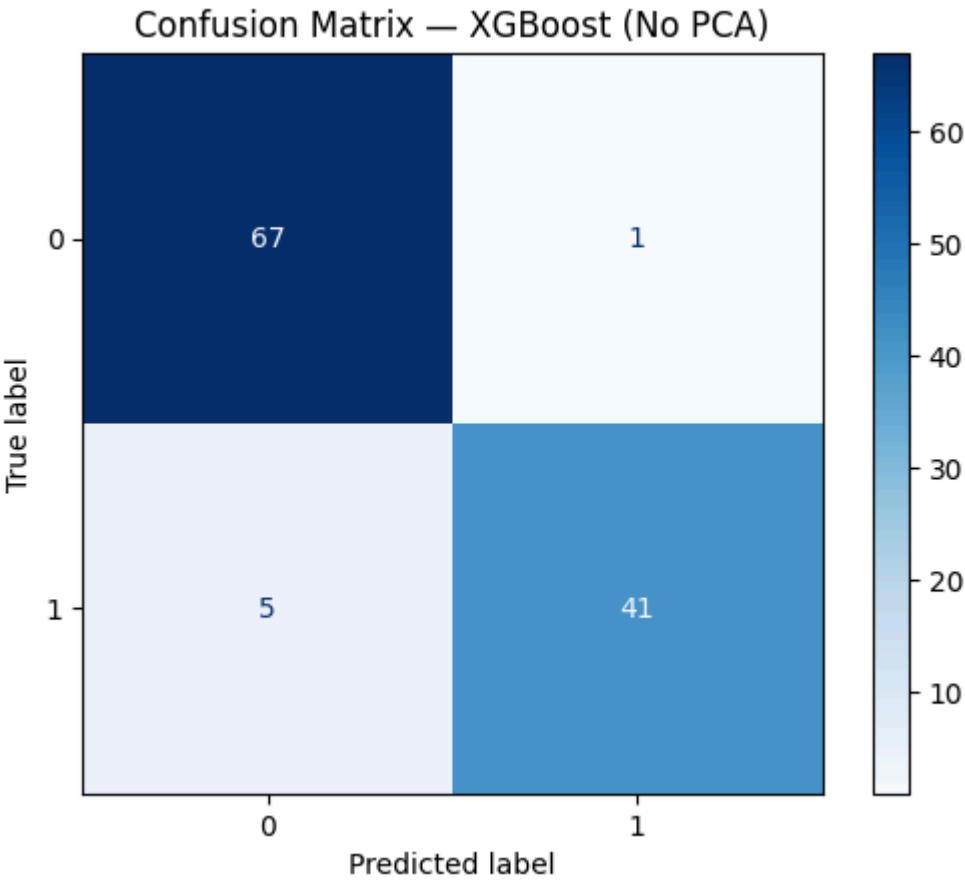


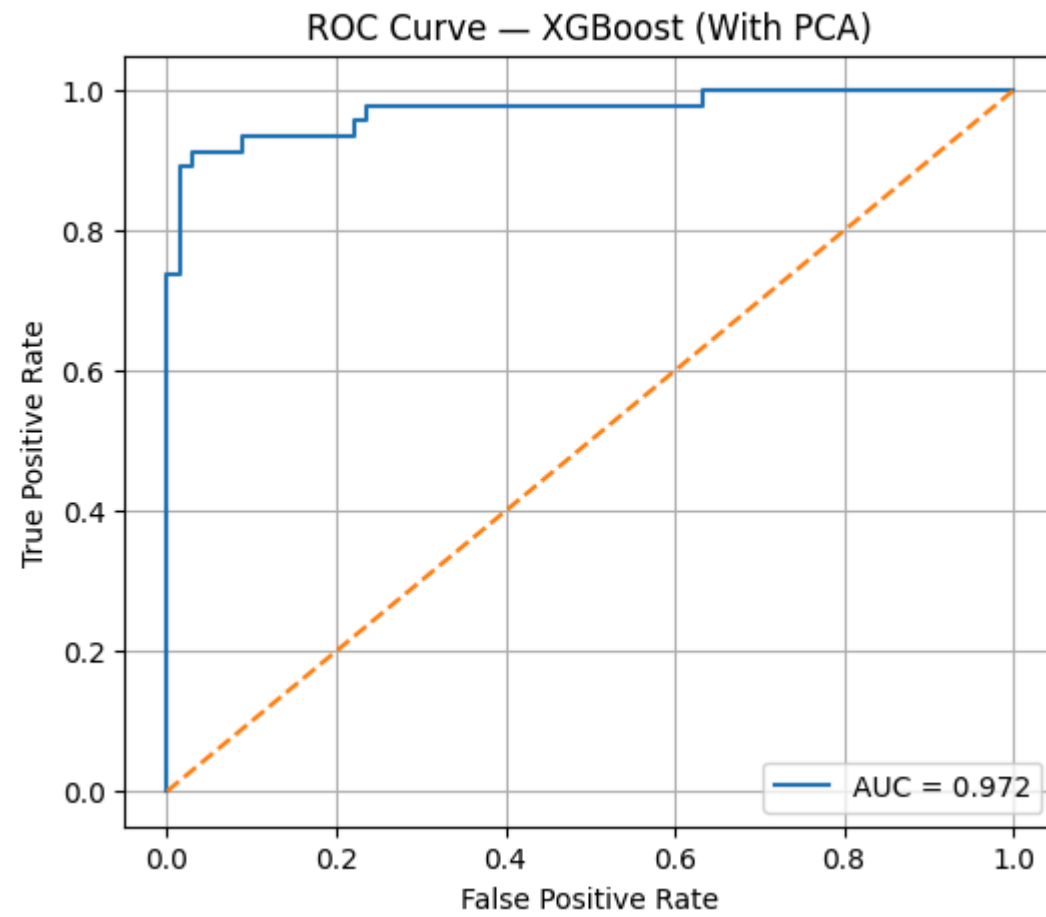


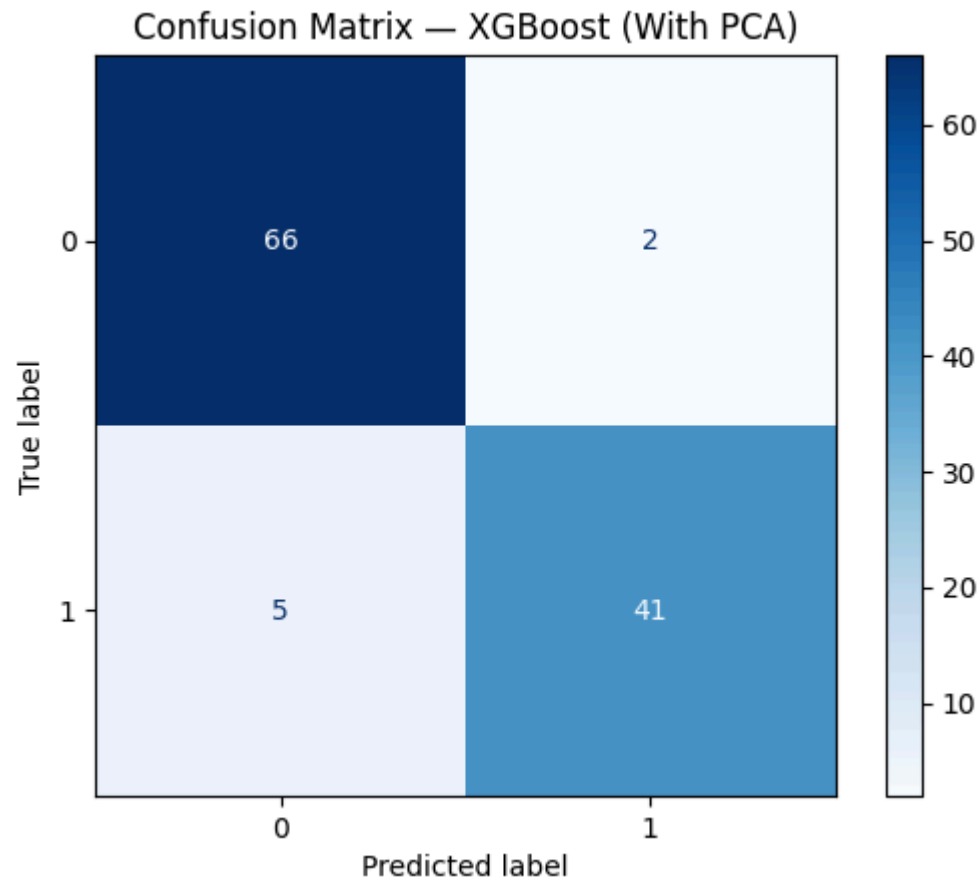


```
In [85]: plot_roc_cm(xgb_no[3], x_test, y_test, "XGBoost (No PCA)")  
plot_roc_cm(xgb_pca[3], x_test_pca, y_test, "XGBoost (With PCA)")
```





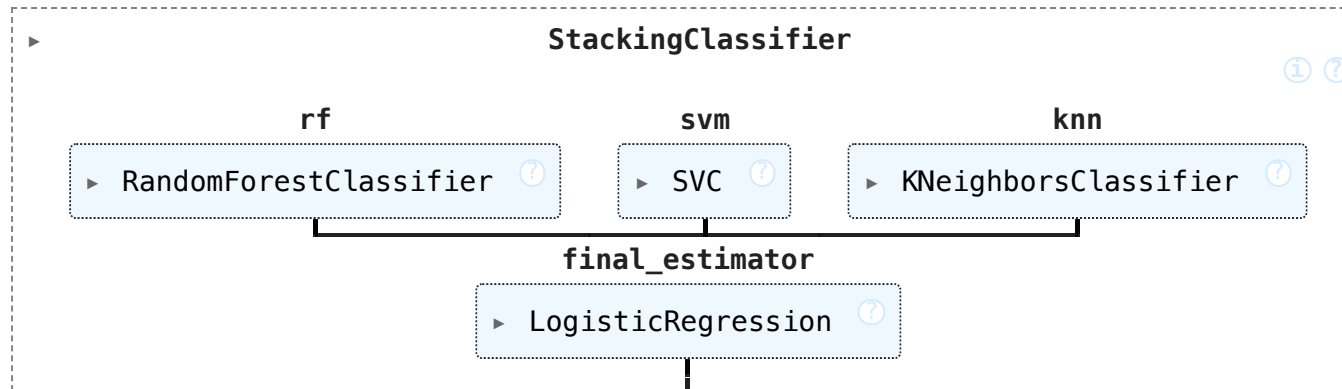




```
In [87]: #Stacking Model
stack_model_no = StackingClassifier(estimators=base_learners, final_estimator=LogisticRegression())
stack_model_no.fit(x_train, y_train)
```

```
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
```

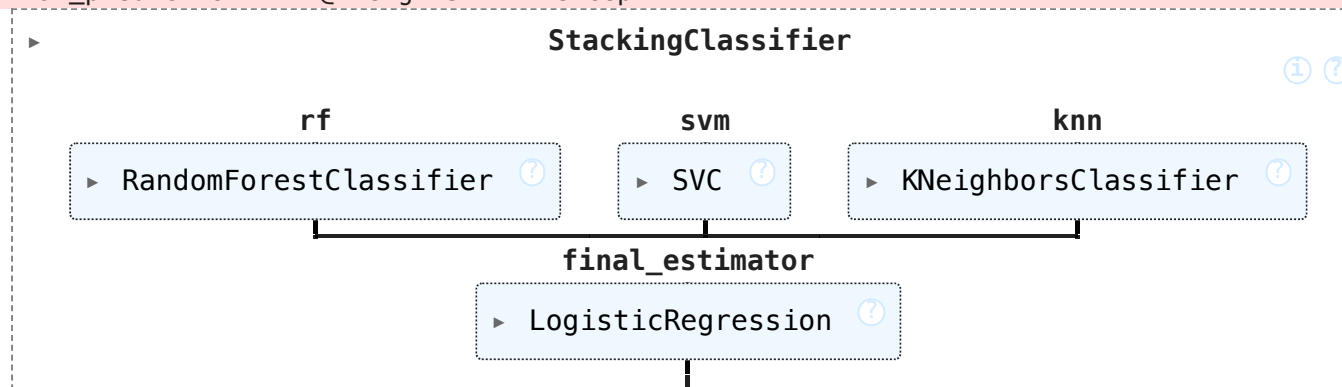
Out [87]:



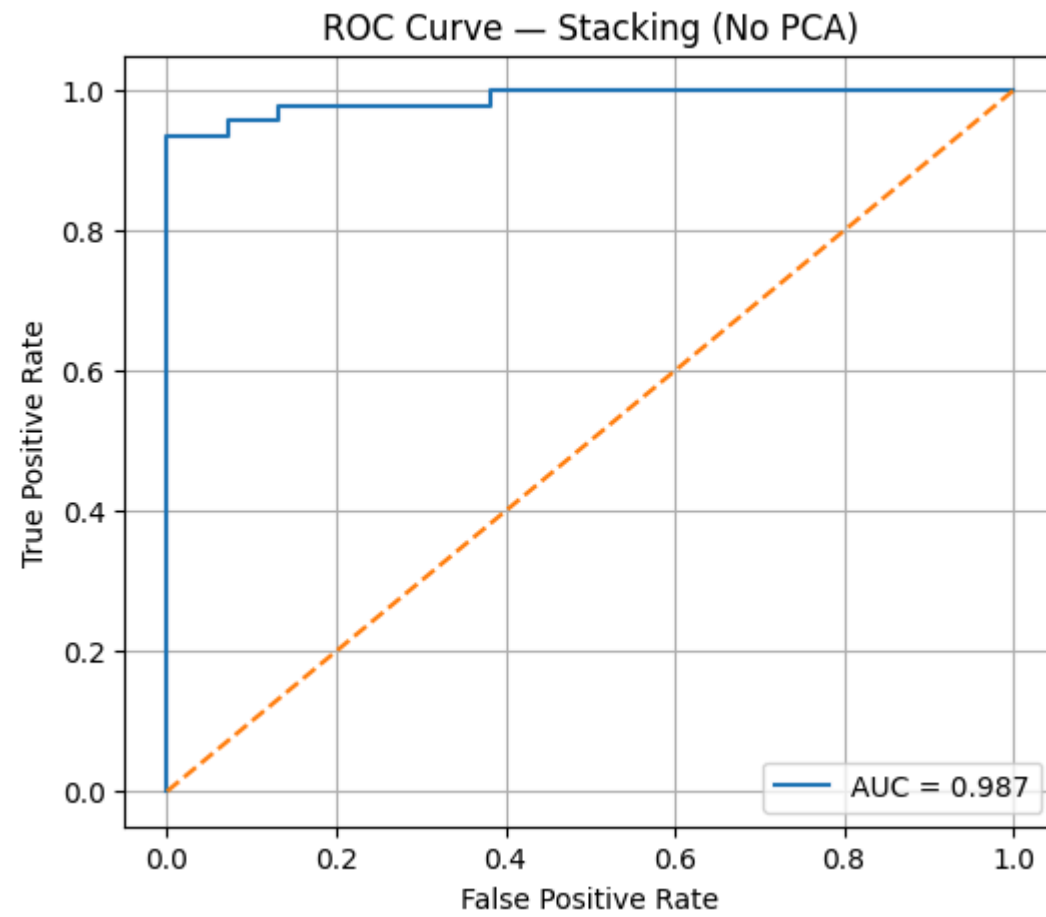
```
In [88]: stack_model_pca = StackingClassifier(estimators=base_learners,final_estimator=LogisticRegression())
stack_model_pca.fit(x_train_pca, y_train)
```

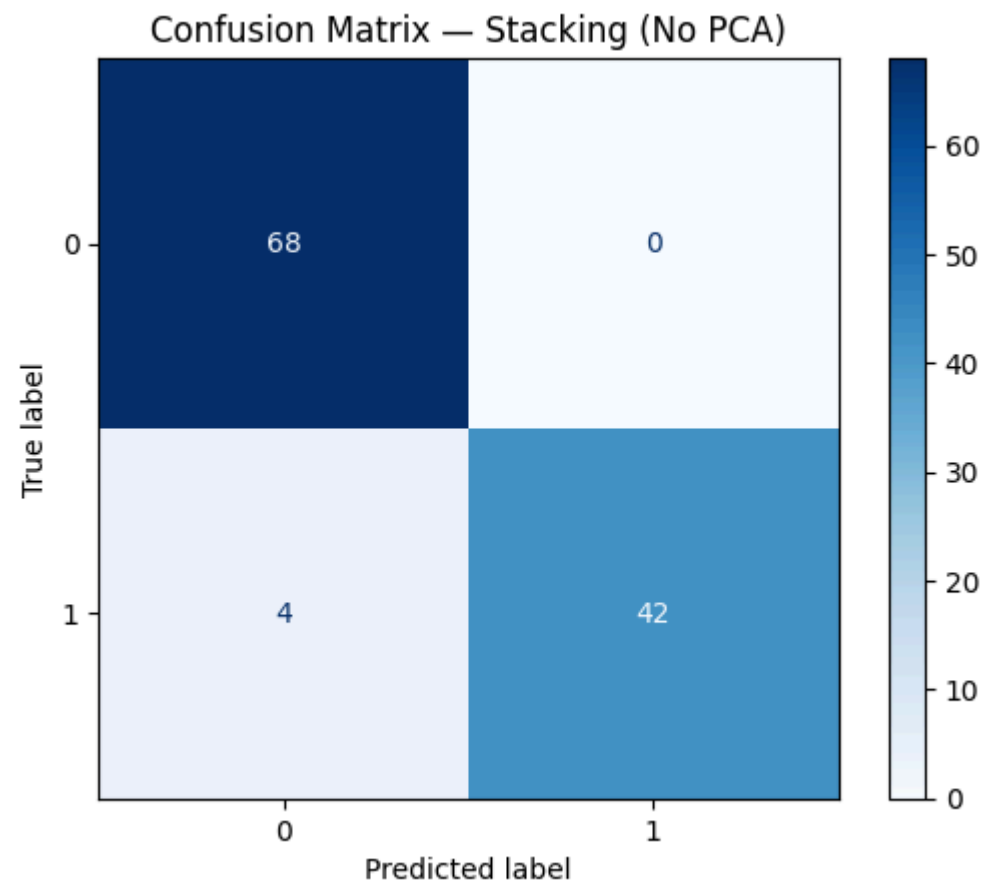
```
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
```

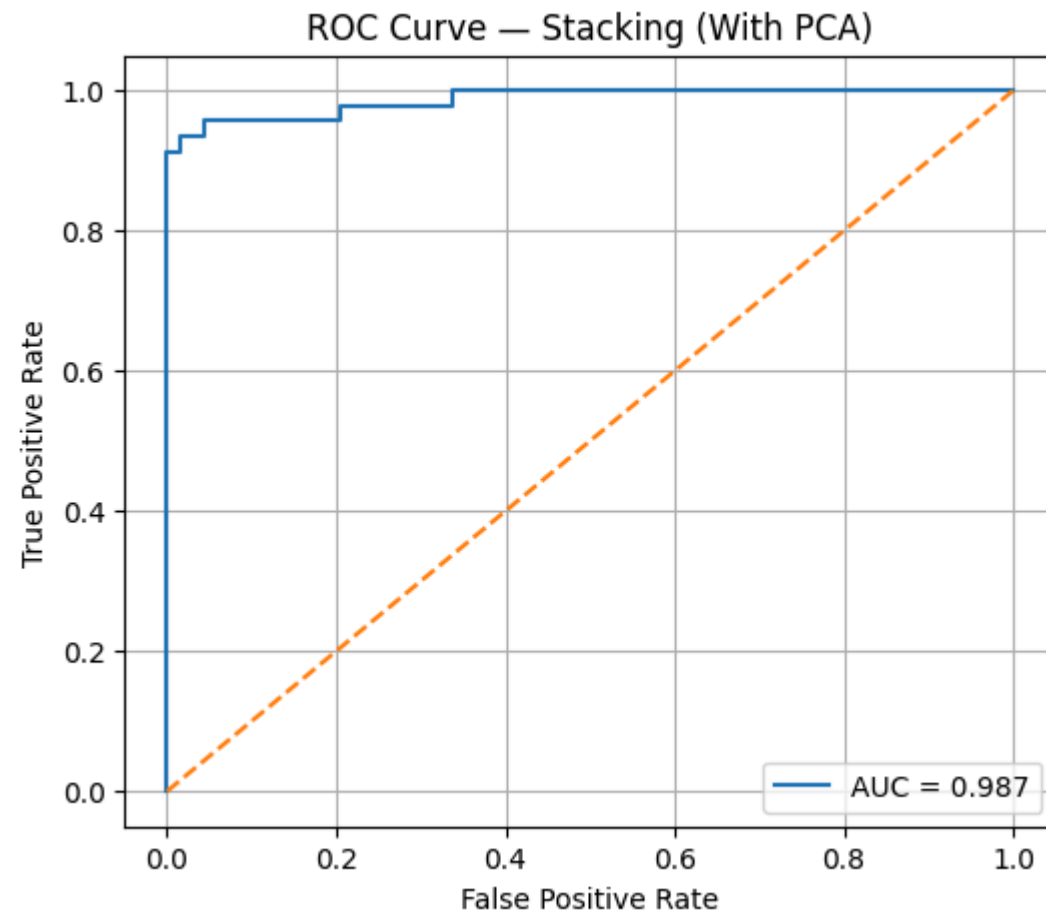
Out [88]:

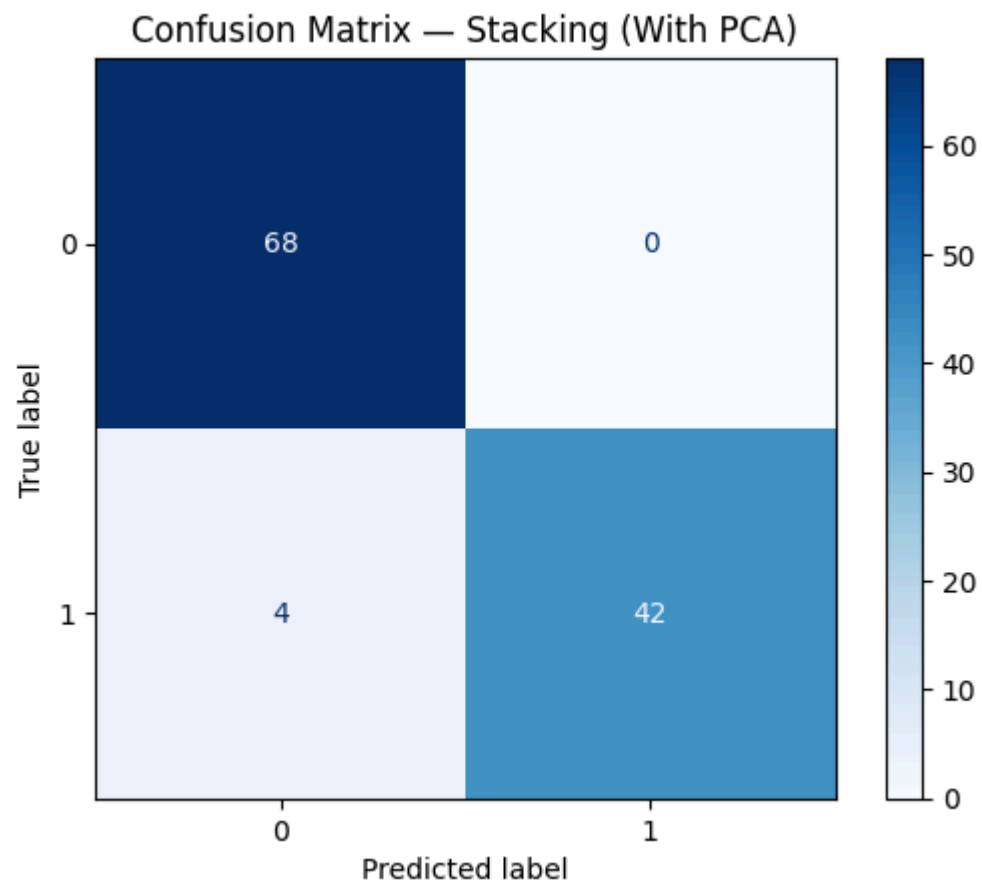


```
In [89]: plot_roc_cm(stack_model_no,x_test,y_test,"Stacking (No PCA)")
plot_roc_cm(stack_model_pca,x_test_pca,y_test,"Stacking (With PCA)")
```









```
In [90]: models_table = pd.DataFrame({"Accuracy (No PCA)": [dt_no[0], rf_no[0], ada_no[0], gb_no[0], xgb_no[0], acc_stack_no],
                                     "F1 (No PCA)": [dt_no[1], rf_no[1], ada_no[1], gb_no[1], xgb_no[1], f1_stack_no],
                                     "Accuracy (With PCA)": [dt_pca[0], rf_pca[0], ada_pca[0], gb_pca[0], xgb_pca[0], acc_st],
                                     "F1 (With PCA)": [dt_pca[1], rf_pca[1], ada_pca[1], gb_pca[1], xgb_pca[1], f1_stack_pca]
                                     })
models_table.round(6)
```

Out [90]:

	Accuracy (No PCA)	F1 (No PCA)	Accuracy (With PCA)	F1 (With PCA)
Decision Tree	0.903509	0.902924	0.929825	0.929553
Random Forest	0.964912	0.964616	0.929825	0.929232
AdaBoost	0.947368	0.946924	0.956140	0.955874
Gradient Boosting	0.964912	0.964616	0.938596	0.937921
XGBoost	0.947368	0.946924	0.938596	0.938224
Stacking	0.964912	0.964616	0.964912	0.964616

```
In [91]: print("Decision Tree Table")
dt_table = pd.DataFrame({
    "Max Depth Tried": ["3, 5, 10"],
    "Min Samples Split Tried": ["2, 5, 10"],
    "Performance (No PCA)": [dt_no[0]],
    "Performance (With PCA)": [dt_pca[0]]
})
dt_table.round(6)
```

Decision Tree Table

Out [91]:

	Max Depth Tried	Min Samples Split Tried	Performance (No PCA)	Performance (With PCA)
0	3, 5, 10	2, 5, 10	0.903509	0.929825

```
In [92]: print("Randomforest Table")
rf_table = pd.DataFrame({
    "Estimators Tried": ["50, 100"],
    "Max Depth Tried": ["5, 10"],
    "Performance (No PCA)": [rf_no[0]],
    "Performance (With PCA)": [rf_pca[0]]
})
rf_table.round(6)
```

Randomforest Table

Out [92]:

	Estimators Tried	Max Depth Tried	Performance (No PCA)	Performance (With PCA)
0	50, 100	5, 10	0.964912	0.929825

```
In [93]: print("AdaBoost Table")
ada_table = pd.DataFrame({
    "Estimators Tried": ["50, 100"],
    "Learning Rate Tried": ["0.5, 1"],
    "Performance (No PCA)": [ada_no[0]],
    "Performance (With PCA)": [ada_pca[0]]
})
ada_table.round(6)
```

AdaBoost Table

Out [93]:

	Estimators Tried	Learning Rate Tried	Performance (No PCA)	Performance (With PCA)
0	50, 100	0.5, 1	0.947368	0.95614

```
In [94]: print("Gradient Boosting Table")
gb_table = pd.DataFrame({
    "Estimators Tried": ["50, 100"],
    "Learning Rate Tried": ["0.05, 0.1"],
    "Performance (No PCA)": [gb_no[0]],
    "Performance (With PCA)": [gb_pca[0]]
})
gb_table.round(6)
```

Gradient Boosting Table

Out [94]:

	Estimators Tried	Learning Rate Tried	Performance (No PCA)	Performance (With PCA)
0	50, 100	0.05, 0.1	0.964912	0.938596

```
In [95]: print("XGBoost Table")
xgb_table = pd.DataFrame({
    "Max Depth Tried": ["3, 5"],
    "Estimators Tried": ["50, 100"],
    "Learning Rate Tried": ["0.05, 0.1"],
    "Performance (No PCA)": [xgb_no[0]],
```

```

    "Performance (With PCA)": [xgb_pca[0]]
})
xgb_table.round(6)

```

XGBoost Table

```

Out [95]:

```

	Max Depth Tried	Estimators Tried	Learning Rate Tried	Performance (No PCA)	Performance (With PCA)
0	3, 5	50, 100	0.05, 0.1	0.947368	0.938596

```

In [96]: print("Stacking Table")
stack_table = pd.DataFrame({
    "Base Learners": ["RF + SVM + KNN"],
    "Meta Learner": ["Logistic Regression"],
    "Performance (No PCA)": [acc_stack_no],
    "Performance (With PCA)": [acc_stack_pca]
})
stack_table.round(6)

```

Stacking Table

```

Out [96]:

```

	Base Learners	Meta Learner	Performance (No PCA)	Performance (With PCA)
0	RF + SVM + KNN	Logistic Regression	0.964912	0.964912

```

In [97]: cv_stack_no = cross_val_score(stack_model, x_train, y_train, cv=5)
cv_stack_pca = cross_val_score(stack_model, x_train_pca, y_train, cv=5)

```

```
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
```

```

erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: di
vide by zero encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: ov
erflow encountered in matmul
    raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: in
valid value encountered in matmul
    raw_prediction = X @ weights + intercept

```

```

/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: divide by zero encountered in matmul
  raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: overflow encountered in matmul
  raw_prediction = X @ weights + intercept
/opt/anaconda3/envs/tf_lab/lib/python3.10/site-packages/sklearn/linear_model/_linear_loss.py:200: RuntimeWarning: invalid value encountered in matmul
  raw_prediction = X @ weights + intercept

```

```

In [98]: cv_table = pd.DataFrame({
    "Fold 1": [
        result_no_pca["SVM"][2][0],
        result_no_pca["Naive Bayes"][2][0],
        result_no_pca["KNN"][2][0],
        cv_lr_no[0],
        dt_no[2][0],
        rf_no[2][0],
        ada_no[2][0],
        gb_no[2][0],
        xgb_no[2][0],
        cv_stack_no[0]],
    "Fold 2": [
        result_no_pca["SVM"][2][1],
        result_no_pca["Naive Bayes"][2][1],
        result_no_pca["KNN"][2][1],
        cv_lr_no[1],
        dt_no[2][1],
        rf_no[2][1],
        ada_no[2][1],
        gb_no[2][1],
        xgb_no[2][1],
        cv_stack_no[1]],
    "Fold 3": [
        result_no_pca["SVM"][2][2],
        result_no_pca["Naive Bayes"][2][2],
        result_no_pca["KNN"][2][2],
        cv_lr_no[2],
        dt_no[2][2],
        rf_no[2][2],
        ada_no[2][2],

```



```
    gb_no[2][2],
    xgb_no[2][2],
    cv_stack_no[2]
],
"Fold 4": [
    result_no_pca["SVM"][2][3],
    result_no_pca["Naive Bayes"][2][3],
    result_no_pca["KNN"][2][3],
    cv_lr_no[3],
    dt_no[2][3],
    rf_no[2][3],
    ada_no[2][3],
    gb_no[2][3],
    xgb_no[2][3],
    cv_stack_no[3]
],
"Fold 5": [
    result_no_pca["SVM"][2][4],
    result_no_pca["Naive Bayes"][2][4],
    result_no_pca["KNN"][2][4],
    cv_lr_no[4],
    dt_no[2][4],
    rf_no[2][4],
    ada_no[2][4],
    gb_no[2][4],
    xgb_no[2][4],
    cv_stack_no[4]
],
"Avg (No PCA)": [
    result_no_pca["SVM"][2].mean(),
    result_no_pca["Naive Bayes"][2].mean(),
    result_no_pca["KNN"][2].mean(),
    cv_lr_no.mean(),
    dt_no[2].mean(),
    rf_no[2].mean(),
    ada_no[2].mean(),
    gb_no[2].mean(),
    xgb_no[2].mean(),
    cv_stack_no.mean()
],
"Avg (With PCA)": [
```

```
    result_pca["SVM"][2].mean(),
    result_pca["Naive Bayes"][2].mean(),
    result_pca["KNN"][2].mean(),
    cv_lr_pca.mean(),
    dt_pca[2].mean(),
    rf_pca[2].mean(),
    ada_pca[2].mean(),
    gb_pca[2].mean(),
    xgb_pca[2].mean(),
    cv_stack_pca.mean()
]
}, index=[
    "SVM",
    "Naive Bayes",
    "KNN",
    "Logistic Regression",
    "Decision Tree",
    "Random Forest",
    "AdaBoost",
    "Gradient Boosting",
    "XGBoost",
    "Stacking"
])
cv_table.round(6)
```

Out [98] :

	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg (No PCA)	Avg (With PCA)
SVM	0.934783	0.902174	0.934783	0.941576	0.932065	0.929076	0.926902
Naive Bayes	0.819293	0.790761	0.808424	0.824728	0.823370	0.813315	0.794565
KNN	0.927989	0.903533	0.907609	0.932065	0.919837	0.918207	0.916848
Logistic Regression	0.921196	0.898098	0.922554	0.933424	0.918478	0.918750	0.916848
Decision Tree	0.967033	0.923077	0.912088	0.868132	0.922222	0.918510	0.938291
Random Forest	0.978022	0.989011	0.956044	0.934066	0.944444	0.960317	0.951502
AdaBoost	0.989011	0.978022	0.956044	0.945055	0.977778	0.969182	0.964786
Gradient Boosting	0.978022	0.978022	0.934066	0.912088	0.955556	0.951551	0.958168
XGBoost	0.967033	0.989011	0.967033	0.945055	0.966667	0.966960	0.973578
Stacking	0.989011	0.978022	0.967033	0.945055	0.955556	0.966935	0.969158

OBSERVATIONS

1. Which models improved most with PCA? Which did not? Why?

Improved most:

Decision Tree, Gradient Boosting, and XGBoost showed improvement in average accuracy with PCA.

Did not improve / slightly decreased:

SVM, Naïve Bayes, KNN, Logistic Regression, Random Forest, and AdaBoost showed slight performance drop.

Reason:

PCA removes feature variance and redundancy. While some ensemble methods benefit from reduced noise, certain models may lose useful discriminative information.

2. Did PCA reduce variance across folds (more stable results)?

Yes. PCA slightly reduced fold variation for Decision Tree and boosting models, indicating improved stability. For linear models, variance change was minimal.

3. For high-dimensional data, was PCA beneficial in reducing overfitting?

Yes. PCA reduced dimensionality and noise, helping models like Decision Tree and boosting methods generalize better, thereby reducing overfitting.

4. How did linear models behave compared to ensemble models with PCA?

Linear models (Logistic Regression, SVM) showed slight performance decrease or remained stable. Ensemble models, especially boosting methods, showed better adaptability and in some cases improved performance with PCA.

5. Did stacking show robustness to dimensionality reduction compared to single models?

Yes. Stacking maintained high performance in both cases, demonstrating robustness to PCA since it combines multiple learners and balances dimensionality effects.

FINAL CONCLUSION

- PCA was beneficial for high-dimensional data with correlated and redundant features.
- It helped reduce noise and overfitting by retaining only important variance.
- Boosting and some ensemble models showed improvement or stable performance after PCA.
- Linear models like Logistic Regression and SVM showed only slight change in performance.
- Tree-based models such as Random Forest and AdaBoost did not benefit much due to built-in feature selection.
- Stacking remained robust in both cases, showing stability to dimensionality reduction.
- Overall, PCA is helpful when redundancy and high dimensionality exist, but less impactful when models already manage feature selection effectively.